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Pyrolysis and combustion chemistry of tetrahydropyran: Experimental and modeling study

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ABSTRACT

This paper reports new experimental and numerical data for the pyrolysis and combustion of tetrahydropyran (THP) – a model component for next-generation heterocyclic oxygenated fuels. Pyrolysis experiments were performed using a plug flow reactor at 170 kPa, over the temperature range 913–1133 K at residence times of approximately 0.5 and 0.2 s, with 90% and 96% N₂ dilution, respectively. THP combustion was investigated in two premixed flat flame burners and in a shock tube. The first premixed flame burner was operated at 6.7 kPa and was used to study detailed flame structures. Two equivalence ratios (1.0 and 1.3) were investigated with a 78% argon dilution. Ethylene, 1,3-butadiene, formaldehyde, and acrolein were the most important intermediates at both pyrolysis and combustion conditions, while the yield of aromatic species was negligible under flame conditions. Laminar burning velocities of THP-air mixtures using the heat flux method were measured at 298, 358 and 398 K and equivalence ratios from 0.55 to 1.50. Finally, ignition delay times of THP-oxygen-argon mixtures were measured behind reflected shock waves at temperatures from 1350 to 1613 K, pressures from 885 to 914 kPa and mixtures containing 0.15–1% fuel at equivalence ratios between 0.5 and 2.0. A new detailed kinetic model for the THP pyrolysis/combustion was developed by EX-GAS, complemented with theoretical calculations for the determining reactions. Good agreements between simulations and acquired experimental data were observed. Reaction path analysis shows that THP is mainly consumed by H-abstractions at both pyrolysis and combustion conditions. The pyrolysis simulations are very sensitive to the unimolecular initiations involving C–C and C–O bond fissions, whereas these reactions play only a very minor role under combustion conditions.

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1. Introduction

Strategies are being developed to selectively transform biomass, containing a variety of substituted five- and six-membered cyclic components, into transportation fuels of cyclic components. Among the main biomass conversion technologies, oxidation and fast pyrolysis of biomass are promising routes for the production of renewable liquid fuels and chemicals. Tetrahydropyran (THP) is a liquid cyclic ether at standard conditions and has a structure (see Fig. 1) that

resembles the core structure of many polysaccharides and sugars, including glucose which is a common raw material for production of biochemicals and biofuels [1–3]. A motored engine study of auto-ignition chemistry of ethers [4] has shown that cyclic ethers have different antiknock characteristics than acyclic ethers, with octane numbers slightly below those of cyclic alkanes with a same ring size. The combustion properties of tetrahydropyran have not yet been fully investigated, but as its molecule includes five carbon atoms, this fuel is expected to have a heating value close to that of gasoline. Due to the presence of an oxygen atom in the molecule, the combustion of tetrahydropyran is expected to form more oxygenated pollutants than hydrocarbon containing fuels. This trend is well reported in previous studies on combustion of oxygenated fuels [5,6]. In addition, THP and its derivatives have been found as important intermediates during the low-temperature oxidation of alkanes. The subsequent reactions of these cyclic ethers are believed to have an effect

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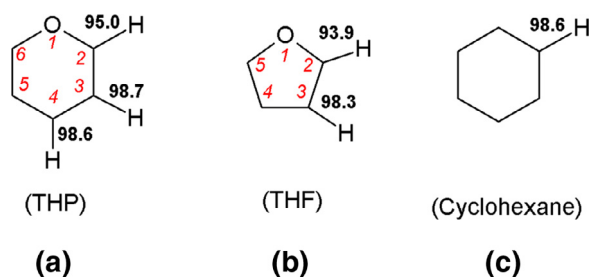


Fig. 1. Structure of THP (a) compared with that of THF (b) [11] and cyclohexane (c) [12]. **Bold** numbers near the C–H bonds are bond energies (in kcal mol⁻¹). *Italic* numbers near the atom are atom labels.

on the overall kinetic mechanism of alkane oxidation [7,8]. In addition to its importance in understanding THP pyrolysis and combustion, the high-temperature chemistry studied in this work is a good starting point to develop future low-temperature models for cyclic ethers.

A better understanding of THP combustion/pyrolysis chemistry is thus useful for improving the fundamental understanding of biomass conversion, alkane oxidation, and the combustion of biofuels having a THP core. However, only very little data regarding the pyrolysis and the combustion of THP is available in literature. To the best of our knowledge, neither the pyrolysis chemistry nor the burning velocity of THP has been studied up to now.

The first THP combustion study dates from 1991 by Leppard. He studied the auto-ignition chemistry of six acyclic and cyclic ethers including THP using a piston engine [4]. The auto-ignition of THP was studied at an equivalence ratio (ϕ) of 0.95 to 1.0, a temperature of 358 K, and a pressure of 60 kPa in the intake manifold. Products formed during the auto-ignition were mainly measured by gas chromatography (GC). The chemical reactions, which are responsible for the auto-ignition of acyclic and cyclic ethers, are used to explain the differences in antiknock characteristics of these ethers. Later, in 1997, ignition delay times and the oxidation of THP were studied by Dagaut et al. [9] behind reflected shock waves ($T = 1000$ – 1700 K, $P = 200$ – 500 kPa, $\phi = 0.5$ – 2) and in a jet-stirred reactor ($T = 800$ – 1100 K, $P = 1000$ kPa, $\phi = 0.5$ – 2), respectively. The chemical species formed during the oxidation of THP were measured by GC. On the basis of these results a detailed kinetic mechanism (72 species, 507 reactions) for the oxidation of THP was proposed. These authors found aldehydes (formaldehyde, acetaldehyde, propanal) to be important intermediates. Most recently, in 2013, Labbe et al. [10] investigated the chemical structure of a fuel-rich premixed low-pressure flame ($P = 2.66$ kPa, $\phi = 1.75$) using vacuum-ultraviolet (VUV)-photoionization molecular-beam mass spectrometry (PI-MBMS). A detailed kinetic model for the oxidation of THP (125 species, 1046 reactions) was developed to interpret the results. Reaction rate coefficients were based on analogy with cyclohexane reactions and theoretical calculations.

In the present work, as part of a continuing effort to enlarge the experimental database for pyrolysis and combustion of oxygenated fuels in general, and to improve the chemical knowledge of THP, a combined *experimental* and *numerical* investigation was carried out. THP pyrolysis was studied in a plug flow reactor and its combustion behavior was studied in three experimental arrangements: a low-pressure premixed flat flame burner (for species profiles), an atmospheric-pressure premixed flat flame adiabatic burner (for laminar burning velocities), and a high-pressure shock tube (for ignition delay times). The results obtained concerning combustion are in line with those previously presented for tetrahydrofuran, using the same experimental set-up and similar conditions [11]. The results are simulated with a newly constructed kinetic model for THP.

2. Experimental methods

The plug flow pyrolysis experiments were carried out at Laboratory for Chemical Technology (LCT), Ghent University, Belgium. The three combustion experiments (two flame burner setups and a shock tube) were performed at the LRGP-CNRS-Nancy, France.

2.1. Tubular flow reactor for pyrolysis

The tubular flow reactor has been used extensively in the past to study the pyrolysis of oxygenated fuels. A detailed description of the setup can be found elsewhere [13–16]. The apparatus consists of a feed, reactor and analysis section.

Liquid THP (Sigma Aldrich, stated purity +99%) is fed to an evaporator kept at 573 K using a peristaltic pump coupled to a Bronkhorst mini-CORI-FLOW Mass Flow to monitor the liquid fuel flow rate. Nitrogen (Air Liquide, purity +99.999%) is used as a diluent and its flow rate is controlled using a Coriolis mass flow controller. Both gasses are mixed prior to entering the reactor.

The reactor has an internal diameter of 6 mm and a length of 1475 mm. It is situated inside of an electrically heated oven which consists of four sections, controlled by four thermocouples. In total, eight thermocouples are positioned along the reactor length, continuously measuring the gas temperature. In the first part of the reactor, the gaseous inlet stream is heated to the set reactor temperature. After approximately 200 mm, the gas temperature reaches the set reactor temperature and remains constant for about 1000 mm. This is followed by a temperature decrease in the final 300 mm of the reactor. An example of gas temperature profiles along the axial reactor coordinate is given in Fig. S1, and all temperature profiles are available as tabulated results in the Supplemental Material. The inlet and outlet pressures are measured by two manometers. The measured pressure drop across the reactor was found to be negligible in all experiments (below 1 kPa).

The reactor effluent was analyzed on-line using two regular gas chromatographs (GC) and one two-dimensional chromatograph (GC×GC). The first gas chromatograph is equipped with two thermal conductivity detectors and one flame ionization detector and is used to quantify all permanent gasses and small hydrocarbons (C_4). The second gas chromatograph, equipped with a single TCD, is dedicated to the quantification of formaldehyde and water. The GC×GC is connected to a time-of-flight mass spectrometer (TOF-MS) and a FID. It permits the identification and quantification of the complete reactor effluent ranging from methane to large polycyclic aromatic hydrocarbons such as pyrene [17,18]. Additional information about the GC settings used in the current study is given in the Supplemental Material. The calibration factors for H_2 , N_2 , CO , CO_2 and C_4 hydrocarbons were determined using a calibration mixture (Air Liquide, Belgium). The response factors of all other species were estimated using the effective carbon number method [19]. N_2 was used as diluent and as internal standard.

Experiments were performed at a constant pressure of 170 kPa and a constant THP molar flow rate of 1.68×10^{-4} mol s⁻¹ over the temperature range of 913–1133 K. Two N_2 dilutions were investigated, i.e. 90 and 96%, corresponding to residence times of approximately 0.5 and 0.2 s (at 1000 K). The pyrolysis conditions are summarized in Table 1.

Triplicate pyrolysis experiments were carried out for all conditions. The obtained results for these triplicates were generally found to be in good agreement. The mole fractions of all species were averaged at each experimental condition. The error was estimated to be 5% for major species, i.e. those with a selectivity higher than 10% (such as THP, CO , H_2 , CH_4 , C_2H_4 , C_2H_6 , and 1,3- C_4H_6), mainly caused by uncertainties in flow rate and calibration factors. Other species, including intermediates, which are present in smaller amounts, have

Table 1

Pyrolysis conditions in plug flow reactor; NL: normal liter.

Gas flow (NL/min)		Dilution ^a (%)	P^b (kPa)	Inlet mass flow rate (g s^{-1})	Set temperature of reactor (K)	Residence time ^c (s)
THP	N ₂					
0.226	2.241	90	170	6.111×10^{-2}	913–1113	0.5
0.226	5.603	96	170	1.311×10^{-1}	913–1133	0.2

^a Dilution = $\text{N}_2/(\text{N}_2 + \text{fuel})$.^b Pressure.^c At 1000 K.**Table 2**

Flow conditions for THP flames; NL: normal liter.

φ^a	Gas flow (NL/min)			Dilution ^b (%)	C/O ratio	P^c (kPa)	Flow velocity at the burner surface ^d (cm s^{-1})	Inlet mass flux ^e ($\text{g cm}^{-2} \text{s}^{-1}$)
	THP (gas)	O ₂	Ar					
1.0	0.187	1.31	4.64	78	0.33	6.7	67	6.41×10^{-3}
1.3	0.241	1.30	4.60	78	0.42	6.7	67	6.48×10^{-3}

^a Equivalence ratio.^b Dilution = $\text{Ar}/(\text{Ar} + \text{O}_2)$.^c Pressure.^d At 333 K^e Mass flow rate per area (of burner surface).

higher uncertainty, approximately on the order of 10%. Elemental balances typically closed within 5%.

2.2. Low-pressure premixed flat flame structure measurements

A complete description of this experimental setup has been given in [11,20–22] and only the main features of the experimental and analytical method used are summarized in this section.

Two premixed flames, one of which is stoichiometric ($\varphi = 1.0$) and the second one is fuel-rich ($\varphi = 1.3$), were investigated at 50 Torr (6.7 kPa) with an argon dilution of 78%. The flames were stabilized on a commercial bronze McKenna burner of 60 mm diameter (water-cooled at 333 K) housed in a vacuum chamber. The burner is mounted on a translational stage, which can be moved in vertical direction.

Flow rates of oxygen (Messer, purity 99.995%) and argon (Messer, purity 99.999%) were controlled by accurate mass flow controllers. Liquid THP (Alfa Aesar, purity >98%) was mixed with argon, then evaporated at 373 K when passing through a Controlled Evaporator-Mixer (CEM) System, and added to the gas stream. Liquid THP was stored in a metallic vessel pressurized with argon. After each load of liquid fuel, argon bubbling and vacuum pumping were performed in order to remove oxygen traces dissolved in the liquid fuel. The flame conditions are summarized in Table 2.

Chemical species are quantified by GC, via a heated (at 423 K) on-line connection to a quartz probe with a 180 μm diameter orifice at the tip and an angle to vertical of 22°, equipped with three types of detectors: a FID coupled with a methanizer, a TCD, and a mass spectrometer (MS). Ar, O₂, H₂ and H₂O were measured by TCD, while other species including CO, CO₂, formaldehyde and all C₁₊ hydrocarbons and oxygenates were detected by FID. In previous work [20], CO and CO₂ could only be detected by TCD, and formaldehyde was not detectable at all. In the current study, CO, CO₂, and formaldehyde were passed through a methanizer in which they were converted to methane, and then detected by FID. The higher sensitivity of a FID reduced the detection limit by a factor of 100 compared to the TCD. Stable species were identified based on their individual retention times and by mass spectrometry (GC/MS). Additional information about the GCs used is given in the Supplemental Material. All detectors were calibrated using cold-gas mixtures. The calibration factors for species that cannot be calibrated directly were estimated, thanks to the hydrogenation in the methanizer, to be the same as that of the alkane involving the same number of carbon atoms. The estimated

uncertainties on the mole fraction measurements of the quantified species were ~5% for the major compounds and ~15% for minor products (<100 ppm). The FID and TCD detection limits were about 1 ppm and 50 ppm, respectively. In all experiments, carbon (C), hydrogen (H), and oxygen (O) atom balances were calculated. The deviation between inlet and outlet is less than 4% for C and ~4–10% for O and H.

Flame temperature profiles were measured with a PtRh (6%)-PtRh (30%) type B thermocouple (diameter 100 μm). The thermocouple wire was supported by an arm and crossed the flame horizontally to avoid conduction heat losses. The thermocouple junction was located at the center of the burner. In order to prevent catalytic effects when placed in a flame, the thermocouple was coated with a ceramic layer of BeO–Y₂O₃ [23] which was obtained by dipping the thermocouple in a hot solution of Y₂(CO₃)₃ (93% mass) and BeO (7% mass) followed by drying in a Meker burner flame. Radiative heat losses were corrected using the electrical compensation method [24]. A sighting telescope (cathetometer) was used to measure the position of the burner relative to the probe or the thermocouple with an accuracy of ± 0.05 mm. The uncertainty of the measured temperature values is estimated to be $\pm 5\%$.

2.3. Laminar burning velocity measurements in atmospheric flat flame

Laminar burning velocities were measured at atmospheric pressure, inlet gas temperatures of 298 K, 358 K, and 398 K and equivalence ratios from 0.55 to 1.50 using the heat flux method [25]. The experimental setup has been described in detail in previous publications [26,27].

The apparatus consists of a brass burner plate (2 mm thick and 30 mm in diameter) mounted on a plenum chamber. The plate is perforated with small holes of diameter 0.5 mm, with the pitch between them being 0.7 mm. Eight type K thermocouples (0.5 mm diameter) are soldered into the plate surface at different distances and angles from the center to the periphery of the burner. The plenum chamber is encompassed by a thermostatic oil jacket, the temperature of which was set to the desired initial temperature of the unburned gas mixture. The circumference of the burner plate was heated with thermostatic oil set to 50 K above the temperature of the unburned gas mixture. In practice, if an initial temperature of 398 K was desired, the temperature of the plenum chamber was set to 398 K and that of the burner plate to 448 K. Thus, the heat gain of the unburned gas mixture can compensate for the heat loss necessary for stabilizing

the flame, knowing that monitoring of the heat loss or gain was performed with the thermocouples.

The error estimation in the laminar burning velocity is described in detail in [26]. The total error of the laminar burning velocity is estimated to be 4%. In addition, the error in the equivalence ratio calculations is about 1%. This results mainly from the error in the mass flow measurements for oxygen and liquid fuel.

2.4. Ignition delay times in shock tube

The stainless-steel shock tube used in the present work was described in detail in [28,29]. It includes a reaction and a driver parts separated by two terphane diaphragms, which were ruptured by suddenly decreasing the pressure in the space separating them. The reaction and the driver parts were respectively 400.6 and 89 cm in length. Helium was used as driver gas. The incident and reflected shock velocities were measured by four piezo-electric pressure transducers located along the reaction section. The pressure and temperature of the test gas behind the reflected shock wave were derived from the values of the initial pressure in the low pressure section and of the incident shock velocity by using ideal one-dimensional shock equations. The error of the temperature measurements is about 20 K. The onset of ignition was detected by the excited OH radical (OH*) emission at 306 nm through a quartz window with a photomultiplier fitted with a monochromator at the end of the reaction part. The quartz window was located at the same axial position as the last pressure transducer. The ignition delay time was defined as the time interval between the pressure rise, measured by the last pressure transducer due to the arrival of the reflected shock wave, and an increase of the photomultiplier signal to 10% of its maximum value (the time resolution of the emission measurements is better than 2 μ s). Fresh reaction mixtures were prepared every day and mixed using a recirculation pump. Before each introduction of the reaction mixture, the reaction section was flushed with pure argon and evacuated to ensure that the residual gas is mainly argon.

The present study was performed at the following experimental conditions, after the reflected shock wave:

- Temperature range of 1350–1613 K and pressure range of 885–914 kPa,
- One fuel-lean ($\varphi = 0.5$) mixture (THP/oxygen/argon, in molar percent): (0.5/7.0/92.5); four stoichiometric ($\varphi = 1.0$) mixtures: (0.15/1.05/98.8), (0.25/1.75/98.0), (0.5/3.5/96.0), (1.0/7.0/92.0); and one fuel-rich ($\varphi = 2.0$) mixture: (0.5/1.75/97.75).

3. Computational methods

3.1. Theoretical rate-constant calculations

The rate coefficients of important THP-related reactions were calculated in two steps. First, the CBS-QB3 method [30] as implemented in the Gaussian 09 revision B suite of programs [31] was used to calculate thermodynamic properties of reactants, products and transition states. The electronic energies were converted with the atomization method to the corresponding heats of formation. B3LYP/6-311G(d,p) level rotational constants and scaled (factor 0.99) harmonic frequencies, except those that represent internal rotations, were used to calculate the entropies, heat capacities and thermal contributions to the enthalpies. Hindrance potentials of internal rotors were determined *via* relaxed scans by increasing the corresponding dihedral angle in 10° steps until complete rotation was achieved. These potentials were fitted to Fourier series prior to their use. The reduced moments of inertia were calculated at the $I^{(2,3)}$ level as defined by East and Radom [32]. With this information, the Schroedinger equation for one-dimensional axis-fixed rotation could be solved and the resulting energy eigenvalues were employed to calculate the contributions of these modes to the thermodynamic functions. Finally, the

thermodynamic data was stored in form of NASA polynomials. In the second step, transition state theory expressed in terms of Gibbs free energies was used to calculate the rate coefficients:

$$k_{\text{TST}}(T) = \chi(T) \cdot \frac{k_B T}{h} \cdot \left(\frac{RT}{p}\right)^{\Delta n - 1} \cdot e^{-\frac{\Delta G^\ddagger}{RT}} \quad (1)$$

$\Delta G^\ddagger$ is the Gibbs free energy difference between transition state without the transitional mode and reactant(s), Δn is the molecularity of the reaction (2 for bimolecular, 1 for unimolecular reactions), and $\chi(T)$ accounts for quantum mechanical tunneling. All other symbols have their usual meaning. The asymmetric Eckart potential was used in this work to calculate $\chi(T)$. The Gibbs free energies were obtained from the NASA polynomials calculated in the first step. Rate coefficients were calculated for the temperature range 300 K–2500 K in 50 K steps and fitted to a modified Arrhenius expression.

3.2. Simulation codes

Simulations of experimental data were performed using CHEMKIN packages [33,34]. The plug flow reactor module (PLUG) was employed for modeling the tubular reactor, supplied with the inlet flow compositions, measured temperature profiles (including temperature increase at the inlet and temperature decrease at the outlet), measured pressure and reactor dimensions. The plug flow assumption for the experimental reactor setup has been verified earlier [13].

Flame structure simulations were performed using the PREMIX module supplied with the inlet flow rate and composition, combustion chamber pressure, and the temperature profile. The average between the experimental profiles measured with and without the sampling probe, shifted 0.6 mm away from the burner surface, was used as temperature profile in the simulations. These corrections are in line with earlier studies [20,21] and account for perturbation effects induced by the sampling quartz probe and the thermocouple: (i) The temperature profile measured without the sampling probe overestimates the temperature while that with the sampling probe underestimates the temperature trajectory of the gas as the probe does not affect the temperature of the gas at all distances above the burner at the same time. The average temperature used here tries to compensate both effects [20,21]; (ii) It has been shown that the probe is actually sampling gases upstream of its physical position [20,35–37]. These probe effects have been thoroughly investigated in [35–37] and a shift equaling 4–5 times the orifice diameter has been recommended for premixed low-pressure flames. The shift of 0.6 mm for temperature profiles used in the present simulations is thus within this upper uncertainty bound of 0.9 mm ($= 5 \times 180 \mu\text{m}$; 180 μm is probe orifice diameter). Simulations using the temperature profiles with upper (shifted 0.9 mm) and lower (shifted 0 mm) uncertainty bounds were also performed. Mole fraction profiles of some primary species of the stoichiometric THP flame are presented in Fig. S2. It can be noted that these shifts do not affect significantly the maximum of simulated mole fraction of intermediate species and products. This observation is in agreement with previous works [11,20,21].

Simulations of adiabatic laminar burning velocities were carried out with the PREMIX module and GRAD and CURV settings of 0.07. Decreasing these parameters slightly decreases simulated burning velocities. However simulations with values lower than 0.07 had difficulty to converge. Thermal diffusion effects are included in the calculations. The mixture-averaged transport approximation was used to reduce the computational burden, with keeping in mind that this is less accurate than the full multicomponent approach. Similarly to the study of Somers et al. [38], the first-order sensitivity coefficient of the mass flow rate to the A-factor of each rate constant was used to identify kinetic parameters controlling the calculated laminar burning velocities.

Shock tube simulations were performed with the SENKIN module. Computed ignition delay times were determined from the 10% increase of the excited OH radical (OH^*) concentration. To calculate OH^* profiles the same sub-mechanism as in the work of Sirjean et al. [39] was used. Note that the experimental OH emission at 306 nm is related to electronically excited OH^* concentration and is not directly proportional to OH radical concentration. Since ignition delay times were determined from the OH^* signal, a sensitivity analysis for this radical was performed to get insight in the reactions that influence the predicted auto-ignition delay times.

4. Kinetic model development

A new detailed chemical kinetic model for the high-temperature combustion and pyrolysis of THP has been developed which includes 273 species and 2031 reactions with primary (reactions 1–93) and secondary (reactions 94–354) mechanisms given in Table 3 and Table S3 (Supplemental Material), respectively. Most of these reactions are written as reversible. However, some reactions are a globalization of several fast successive elementary steps. They need then to be written as irreversible since the reverse rate constant cannot be calculated from the global equilibrium constant. In this category are for instance THP unimolecular initiations through diradical intermediates or succession of H-abstractions/ β -scissions for dihydropyrans as described in the following parts. The structure of the newly involved species can be found in Table S4 in the Supplemental Material. The model was developed in two steps:

- (i) *In the first step*, a kinetic mechanism, including thermochemical and transport data, for THP combustion was generated by EXGAS [40–44] starting from a C_0 – C_6 reactions base.
- (ii) *In the second step*, some key kinetic parameters in the automatically generated mechanism were updated. The kinetic parameters provided by EXGAS are mainly derived and validated for alkanes and alkenes and, hence, they do not take into account the specificity of cyclic ether reactivity. Updated kinetic parameters were taken from quantum calculation, as presented above, Evans–Polanyi correlations or analogy with structurally similar species. Both steps will be discussed in the following paragraphs.

4.1. Mechanism generated with EXGAS

The kinetic model generated with the new EXGAS software consists of three parts:

- (i) The first part is a C_0 – C_6 reactions base mechanism [44], which consists of a C_0 – C_2 sub-mechanism [40,42] with associated kinetic data mainly taken from [45,46], reactions of C_3 – C_6 species [47–49], and reactions of small aromatic compounds up to ethylbenzene [44]. Recently [11], reaction pathways involving fulvene and leading to benzene formation have been added as proposed by Li et al. [50].
- (ii) The second part is a comprehensive *primary mechanism* of the THP high-temperature oxidation containing fuel consumption reactions, i.e. unimolecular initiation by C–H bond fissions (reactions 6–8), bimolecular initiations of THP with O_2 (reactions 9–11) and H-abstractions (reactions 12–47) and THP radical decomposition reactions, i.e. β - and α -scissions (reactions 48–79), oxidations and isomerizations (reactions 80–93). Note that, the unimolecular initiations of THP by C–O and C–C bond fissions (reactions 1–5) cannot be generated by EXGAS. This part of the *primary mechanism* will be discussed separately below.
- (iii) The third part is a global *secondary mechanism* that contains the reactions of the molecular products formed by the *primary*

mechanism. They are already included in the C_0 – C_6 reactions base, or proposed by EXGAS.

4.2. Kinetic parameters in the primary mechanism of THP

Some key kinetic parameters in the *primary mechanism* of THP were updated with quantum calculations, Evans–Polanyi correlations, and analogy with similar structured species.

4.2.1. Unimolecular decomposition

Recently, Verdicchio et al. [51] performed a detailed study of the unimolecular decomposition of tetrahydrofuran (THF) by means of high-level quantum chemistry calculations. Five types of unimolecular initiation pathways were examined for the THF decomposition: (i) initial C–H bond fissions involving CH_2 moieties of the THF-ring, (ii) ring-opening by C–O and C–C bond fissions yielding C_4 unsaturated ethers and n-butanal via diradical species, (iii) a pericyclic rearrangement to yield an unsaturated alcohol (1-buten-4-ol), (iv) ring-opening by C–O and C–C bond fission with internal hydrogen atom transfer leading to the formation of carbenic intermediates that isomerize and decompose into molecular products, and (v) a pericyclic hydrogen elimination involving the formation of cyclic diradical. Both THF and THP are saturated cyclic ethers and, thus, their unimolecular decomposition can assumed to be similar as a first approximation. However, the difference in ring size (5 atoms in THF, 6 atoms in THP; see Fig. 1) will obviously result in different ring strains. This will have an effect on rate coefficients and branching ratios of their respective decomposition pathways. Therefore, one cannot directly use the rate coefficients from the THF calculations by Verdicchio et al. [51] for THP. Considering the ring corrections to the group additivity method of Benson for thermodynamic properties, it can be noted that the contributions of the THP ring strain to $\Delta_f H^\circ$ (298 K) and S° (298 K) are around 0.5 kcal mol^{−1} and around 18.8 cal mol^{−1} K^{−1}, respectively, and are very different to those of the THF ring strain, i.e., 5.9 kcal mol^{−1} to $\Delta_f H^\circ$ (298 K) and 25.3 cal mol^{−1} K^{−1} to S° (298 K) [52]. However, the contributions of THP ring strain are very close to those of the cyclohexane ring, i.e. 0 kcal mol^{−1} to $\Delta_f H^\circ$ (298 K) and around 18.8 cal mol^{−1} K^{−1} to S° (298 K) [52]. These values are summarized in Table 4. Like THP, cyclohexane is a saturated ring molecule with 6 heavy atoms in the ring. Since a complete set of kinetic data for the unimolecular reactions of THP (C–H bond breaking, pericyclic, diradical, and carbenic mechanisms) is unavailable, most of the required rate constants were estimated by analogy between cyclohexane and THP. One objective of this study is to determine the importance of these reactions within the THP mechanism. Should the performance of the model strongly depend on these reactions, future research would be dedicated to study the unimolecular chemistry of THP in detail with first principle methods.

The rate constants for ring-opening by C–O and C–C bond scission yielding either n-pentanal (CCCCCHO) (reaction 2), or one of the two possible unsaturated ethers (CCCO^{*}C and C^{*}CCOC; reactions 3 and 4) via diradicals, were estimated by analogy with cyclohexane [53]. Unlike cyclohexane, there are three different distinct carbon sites on THP (Fig. 1): C_2 and C_6 (carbons directly adjacent to the ether oxygen); C_3 and C_5 (the next neighboring two carbons); and C_4 (carbon opposite the oxygen). Therefore, the activation energies were modified considering (i) the difference in BDE between C–C and C–O bonds in THP molecule and (ii) the difference between ring strain between THP and cyclohexane.

As the α -carbene (produced from ring-opening by C_1 – C_2 bond fission) is a relatively stable intermediate in the cyclic ether chemistry [51], its formation pathway from THP has been considered in the present model. This α -carbene (structure in Table S4) can decompose to 1-butene and formaldehyde (reaction 5). The kinetic parameters of this reaction were taken from the THF calculations [11,51]

Table 3

Primary mechanism of the high-temperature pyrolysis and oxidation of THP. The rate constants are given ($k = AT^n \exp(-E_a/RT)$) in cm^3 , mol, s, cal units. The detailed structures of the newly involved species can be found in Table S4 in the Supplemental Material.

No.	Reactions Primary mechanism	A	n	E_a	Footnote
Reactions of THP					
Unimolecular initiations					
1	$\text{THP} \rightarrow \text{C}^* \text{CCCCOH}$	4.20×10^5	2.6	73300.0	e, #
2	$\text{THP} \rightarrow \text{CCCCCHO}$	6.50×10^{19}	−0.6	90120.0	a
3	$\text{THP} \rightarrow \text{CCCO}^* \text{C}$	6.50×10^{19}	−0.6	94020.0	a
4	$\text{THP} \rightarrow \text{C}^* \text{CCCOC}$	6.50×10^{19}	−0.6	91420.0	a
5	$\text{THP} \rightarrow 1\text{-C}_4\text{H}_8 + \text{HCHO}$	1.08×10^{26}	−3.0	95111.0	b
6	$\text{THP} \rightleftharpoons \text{H} + \text{THP-YL-3}$	4.90×10^{14}	0.0	98798.8	c
7	$\text{THP} \rightleftharpoons \text{H} + \text{THP-YL-2}$	4.80×10^{14}	0.0	95079.8	c
8	$\text{THP} \rightleftharpoons \text{H} + \text{THP-YL-4}$	4.90×10^{14}	0.0	98698.8	c
Bimolecular initiations					
9	$\text{THP} + \text{O}_2 \rightleftharpoons \text{THP-YL-3} + \text{OOH}$	4.00×10^{13}	0.0	50749.2	c
10	$\text{THP} + \text{O}_2 \rightleftharpoons \text{THP-YL-2} + \text{OOH}$	4.00×10^{13}	0.0	47021.9	c
11	$\text{THP} + \text{O}_2 \rightleftharpoons \text{THP-YL-4} + \text{OOH}$	2.00×10^{13}	0.0	50649.2	c
H-abstractions					
12	$\text{THP} + \text{O} \rightleftharpoons \text{THP-YL-2} + \text{OH}$	6.80×10^8	1.5	1180.0	d
13	$\text{THP} + \text{O} \rightleftharpoons \text{THP-YL-3} + \text{OH}$	6.80×10^8	1.5	3932.5	d
14	$\text{THP} + \text{O} \rightleftharpoons \text{THP-YL-4} + \text{OH}$	3.40×10^8	1.5	3857.5	d
15	$\text{THP} + \text{H} \rightarrow \text{THP-YL-2} + \text{H}_2$	2.50×10^6	2.4	4000.0	e, #
16	$\text{THP} + \text{H} \rightarrow \text{THP-YL-3} + \text{H}_2$	3.70×10^7	2.0	6800.0	e, #
17	$\text{THP} + \text{H} \rightarrow \text{THP-YL-4} + \text{H}_2$	2.10×10^7	2.0	6000.0	e, #
18	$\text{THP} + \text{OH} \rightleftharpoons \text{THP-YL-2} + \text{H}_2\text{O}$	4.80×10^6	2.0	−2095.0	d
19	$\text{THP} + \text{OH} \rightleftharpoons \text{THP-YL-3} + \text{H}_2\text{O}$	4.80×10^6	2.0	−260.0	d
20	$\text{THP} + \text{OH} \rightleftharpoons \text{THP-YL-4} + \text{H}_2\text{O}$	2.40×10^6	2.0	−310.0	d
21	$\text{THP} + \text{OOH} \rightleftharpoons \text{THP-YL-2} + \text{H}_2\text{O}_2$	5.60×10^4	2.7	15,990.0	d
22	$\text{THP} + \text{OOH} \rightleftharpoons \text{THP-YL-3} + \text{H}_2\text{O}_2$	5.60×10^4	2.7	18,192.0	d
23	$\text{THP} + \text{OOH} \rightleftharpoons \text{THP-YL-4} + \text{H}_2\text{O}_2$	2.80×10^4	2.7	18,132.0	d
24	$\text{THP} + \text{CH}_3 \rightarrow \text{THP-YL-2} + \text{CH}_4$	2.00×10^2	3.2	7600.0	e, #
25	$\text{THP} + \text{CH}_3 \rightarrow \text{THP-YL-3} + \text{CH}_4$	9.00×10^2	3.0	9000.0	e, #
26	$\text{THP} + \text{CH}_3 \rightarrow \text{THP-YL-4} + \text{CH}_4$	5.60×10^2	3.0	8700.0	e, #
27	$\text{THP} + \text{C}_2\text{H}_3 \rightleftharpoons \text{THP-YL-2} + \text{C}_2\text{H}_4$	4.00×10^{11}	0.0	11,000.0	f
28	$\text{THP} + \text{C}_2\text{H}_3 \rightleftharpoons \text{THP-YL-3} + \text{C}_2\text{H}_4$	4.00×10^{11}	0.0	11,000.0	f
29	$\text{THP} + \text{C}_2\text{H}_3 \rightleftharpoons \text{THP-YL-4} + \text{C}_2\text{H}_4$	2.00×10^{11}	0.0	11,000.0	f
30	$\text{THP} + \text{C}_2\text{H}_5 \rightleftharpoons \text{THP-YL-2} + \text{C}_2\text{H}_6$	4.00×10^{11}	0.0	11,000.0	c
31	$\text{THP} + \text{C}_2\text{H}_5 \rightleftharpoons \text{THP-YL-3} + \text{C}_2\text{H}_6$	4.00×10^{11}	0.0	11,000.0	c
32	$\text{THP} + \text{C}_2\text{H}_5 \rightleftharpoons \text{THP-YL-4} + \text{C}_2\text{H}_6$	2.00×10^{11}	0.0	11,000.0	c
33	$\text{THP} + \text{C}_3\text{H}_5\text{Y} \rightarrow \text{THP-YL-2} + \text{C}_3\text{H}_6$	3.70×10^1	3.6	17,500.0	e, #
34	$\text{THP} + \text{C}_3\text{H}_5\text{Y} \rightarrow \text{THP-YL-3} + \text{C}_3\text{H}_6$	1.20×10^3	3.2	19,900.0	e, #
35	$\text{THP} + \text{C}_3\text{H}_5\text{Y} \rightarrow \text{THP-YL-4} + \text{C}_3\text{H}_6$	1.20×10^3	3.1	19,300.0	e, #
36	$\text{THP} + \text{CHO} \rightleftharpoons \text{THP-YL-2} + \text{HCHO}$	2.20×10^7	1.9	17,000.0	c
37	$\text{THP} + \text{CHO} \rightleftharpoons \text{THP-YL-3} + \text{HCHO}$	2.20×10^7	1.9	17,000.0	c
38	$\text{THP} + \text{CHO} \rightleftharpoons \text{THP-YL-4} + \text{HCHO}$	1.00×10^7	1.9	17,000.0	c
39	$\text{THP} + \text{CH}_2\text{OH} \rightleftharpoons \text{THP-YL-2} + \text{CH}_3\text{OH}$	1.20×10^2	3.0	12,000.0	c
40	$\text{THP} + \text{CH}_2\text{OH} \rightleftharpoons \text{THP-YL-3} + \text{CH}_3\text{OH}$	1.20×10^2	3.0	12,000.0	c
41	$\text{THP} + \text{CH}_2\text{OH} \rightleftharpoons \text{THP-YL-4} + \text{CH}_3\text{OH}$	6.00×10^1	3.0	12,000.0	c
42	$\text{THP} + \text{CH}_3\text{O} \rightleftharpoons \text{THP-YL-2} + \text{CH}_3\text{OH}$	2.90×10^{11}	0.0	4500.0	c
43	$\text{THP} + \text{CH}_3\text{O} \rightleftharpoons \text{THP-YL-3} + \text{CH}_3\text{OH}$	2.90×10^{11}	0.0	4500.0	c
44	$\text{THP} + \text{CH}_3\text{O} \rightleftharpoons \text{THP-YL-4} + \text{CH}_3\text{OH}$	1.50×10^{11}	0.0	4500.0	c
45	$\text{THP} + \text{CH}_3\text{OO} \rightleftharpoons \text{THP-YL-2} + \text{CH}_3\text{OOH}$	6.00×10^{12}	0.0	17,500.0	c
46	$\text{THP} + \text{CH}_3\text{OO} \rightleftharpoons \text{THP-YL-3} + \text{CH}_3\text{OOH}$	6.00×10^{12}	0.0	17,500.0	c
47	$\text{THP} + \text{CH}_3\text{OO} \rightleftharpoons \text{THP-YL-4} + \text{CH}_3\text{OOH}$	3.00×10^{12}	0.0	17,500.0	c
Reactions of THP-YL-3, THP-YL-2, THP-YL-4 and subsequently derived radicals					
Ring-openings					
48	$\text{THP-YL-2} \rightarrow \text{R52C}_5\text{H}_9\text{OA}$	5.80×10^{11}	0.6	23,100.0	e, #
49	$\text{THP-YL-2} \rightarrow \text{R53C}_5\text{H}_9\text{OEZ}$	1.00×10^{12}	0.8	33,600.0	e, #
50	$\text{THP-YL-3} \rightarrow \text{R50C}_5\text{H}_9\text{OEZ}$	1.50×10^{12}	0.6	30,500.0	e, #
51	$\text{THP-YL-3} \rightarrow \text{R51C}_5\text{H}_9\text{OZ}$	4.80×10^{12}	0.3	21,900.0	e, #
52	$\text{THP-YL-4} \rightarrow \text{R54C}_5\text{H}_9\text{OEZ}$	2.00×10^{12}	0.5	28,100.0	e, #
β -Scissions					
53	$\text{THP-YL-2} \rightarrow 3,4\text{-DHP} + \text{H}$	2.80×10^4	2.7	45,675.2	b, #
54	$\text{THP-YL-3} \rightarrow 3,4\text{-DHP} + \text{H}$	1.20×10^{11}	0.9	30,700.0	e, #
55	$\text{THP-YL-3} \rightarrow 3,6\text{-DHP} + \text{H}$	1.40×10^{11}	0.9	34,100.0	e, #
56	$\text{THP-YL-4} \rightarrow 3,6\text{-DHP} + \text{H}$	1.40×10^{11}	0.9	35,900.0	e, #
57	$\text{R50C}_5\text{H}_9\text{OEZ} \rightarrow \text{R56C}_3\text{H}_5\text{OZ} + \text{C}_2\text{H}_4$	2.60×10^{10}	0.9	23,000.0	e, #
58	$\text{R51C}_5\text{H}_9\text{OZ} \rightarrow \text{C}_4\text{H}_7\text{-1} + \text{HCHO}$	1.30×10^{14}	0.0	13,700.0	e, #
59	$\text{R52C}_5\text{H}_9\text{OA} \rightarrow \text{R60C}_3\text{H}_5\text{OA} + \text{C}_2\text{H}_4$	2.00×10^{11}	0.7	28,600.0	e, #
60	$\text{R53C}_5\text{H}_9\text{OEZ} \rightarrow \text{R62C}_3\text{H}_5\text{OEZ} + \text{C}_2\text{H}_4$	2.40×10^{12}	0.4	27,800.0	e, #
61	$\text{R54C}_5\text{H}_9\text{OEZ} \rightarrow \text{C}_4\text{H}_7\text{-1} + \text{HCHO}$	6.40×10^{12}	0.4	24,400.0	e, #

(continued on next page)

Table 3 (continued)

No.	Reactions Primary mechanism	A	n	E _a	Footnote
62	R55C ₅ H ₉ OEX ⇒ C ₂ H ₅ + C ₂ H ₃ CHO	4.30 × 10 ¹²	0.0	33,200.0	^c
63	R56C ₃ H ₅ OZ ⇒ C ₂ H ₃ + HCHO	9.70 × 10 ¹⁵	−1.1	20,640.0	^c
64	R57C ₅ H ₉ OA ⇒ CHO + 1-C ₄ H ₈	2.00 × 10 ¹³	0.0	27,700.0	^c
65	R57C ₅ H ₉ OA ⇒ CH ₃ + C ₄ H ₆ OAZ	2.00 × 10 ¹³	0.0	31,000.0	^c
66	R58C ₅ H ₉ OA ⇒ C ₂ H ₅ + C ₂ H ₃ CHO	2.00 × 10 ¹³	0.0	28,700.0	^c
67	R60C ₃ H ₅ OA ⇒ CHO + C ₂ H ₄	2.00 × 10 ¹³	0.0	27,700.0	^c
68	R61C ₅ H ₉ OZ ⇒ C ₂ H ₃ + C ₂ H ₅ CHO	2.00 × 10 ¹³	0.0	24,000.0	^c
69	R61C ₅ H ₉ OZ ⇒ CH ₃ + C ₄ H ₆ OZ2	2.00 × 10 ¹³	0.0	31,000.0	^c
70	R62C ₃ H ₅ OZ ⇒ C ₂ H ₃ + HCHO	2.00 × 10 ¹³	0.0	24,000.0	^c
71	R63C ₅ H ₉ OZ ⇒ CH ₃ + C ₄ H ₆ OAZ	2.00 × 10 ¹³	0.0	24,000.0	^c
72	R63C ₅ H ₉ OZ ⇒ C ₂ H ₃ + C ₃ H ₆ OZ	2.00 × 10 ¹³	0.0	35,500.0	^c
73	R64C ₅ H ₉ OEX ⇒ CH ₃ O + 1,3-C ₄ H ₆	4.30 × 10 ¹²	0.0	33,200.0	^c
74	R65C ₅ H ₉ OZ ⇒ C ₃ H ₅ Y + CH ₃ CHO	2.00 × 10 ¹³	0.0	24,000.0	^c
75	R66C ₅ H ₉ OA ⇒ CH ₂ CHO + C ₃ H ₆	2.00 × 10 ¹³	0.0	28,700.0	^c
76	nC ₄ H ₉ ⇒ C ₂ H ₅ + C ₂ H ₄	2.00 × 10 ¹³	0.0	28,700.0	^c
77	C ₄ H ₉ -2 ⇒ CH ₃ + C ₃ H ₆	2.00 × 10 ¹³	0.0	31,000.0	^c
<i>α</i> -Scissions					
78	R59C ₅ H ₉ OK ⇒ nC ₄ H ₉ + CO	2.80 × 10 ¹³	0.0	17,150.0	^c
79	R67C ₃ H ₅ OK ⇒ C ₂ H ₅ + CO	2.80 × 10 ¹³	0.0	17,150.0	^c
Oxidations					
80	R55C ₅ H ₉ OEX + O ₂ ⇒ C ₅ H ₈ OZ2 + OOH	1.00 × 10 ¹²	0.0	22,730.0	^c
81	R64C ₅ H ₉ OEX + O ₂ ⇒ C ₅ H ₈ OZ2 + OOH	1.00 × 10 ¹²	0.0	22,730.0	^c
Isomerizations					
82	nC ₄ H ₉ ⇒ C ₄ H ₉ -2	3.30 × 10 ⁹	1.0	37,000.0	^c
83	R50C ₅ H ₉ OZ ⇒ R55C ₅ H ₉ OEX	5.70 × 10 ⁸	1.0	16,940.0	^c
84	R52C ₅ H ₉ OA ⇒ R57C ₅ H ₉ OA	3.30 × 10 ⁹	1.0	37,000.0	^c
85	R52C ₅ H ₉ OA ⇒ R58C ₅ H ₉ OA	5.70 × 10 ⁸	1.0	17,300.0	^c
86	R52C ₅ H ₉ OA ⇒ R59C ₅ H ₉ OK	4.90 × 10 ⁷	1.0	8000.0	^c
87	R53C ₅ H ₉ OZ ⇒ R61C ₅ H ₉ OZ	3.30 × 10 ⁹	1.0	36,640.0	^c
88	R54C ₅ H ₉ OZ ⇒ R63C ₅ H ₉ OZ	3.30 × 10 ⁹	1.0	42,640.0	^c
89	R54C ₅ H ₉ OZ ⇒ R64C ₅ H ₉ OEX	5.70 × 10 ⁸	1.0	24,940.0	^c
90	R55C ₅ H ₉ OEX ⇒ R65C ₅ H ₉ OZ	3.30 × 10 ⁹	1.0	46,500.0	^c
91	R57C ₅ H ₉ OA ⇒ R59C ₅ H ₉ OK	1.70 × 10 ⁹	1.0	33,000.0	^c
92	R58C ₅ H ₉ OA ⇒ R66C ₅ H ₉ OA	3.30 × 10 ⁹	1.0	37,000.0	^c
93	R60C ₃ H ₅ OA ⇒ R67C ₃ H ₅ OK	1.70 × 10 ⁹	1.0	33000.0	^c

^a Analogy with the reactions of cyclohexane of Sirjean et al. [53]. Difference in BDE between C–C and C–O in THP molecule, and difference between ring strain of THP and that of cyclohexane have been taken into account.

^b Analogy with the reactions of THF of Verdicchio et al. [51].

^c Directly provided by EXGAS software [42].

^d Estimated using the Evans–Polanyi correlation proposed by Dean and Bozelli [54].

^e Calculated using CBS–QB3 methods.

^f Rate constant taken equal to that of H-abstractions by C₂H₅ as provided by EXGAS software.

^g The reverse reaction is given in the complete mechanism in the Supplemental Material.

without any further modifications, even though this creates a rather large uncertainty. Given that THF contains more ring strain than THP, the used activation energy should be a lower limit for THP and consequently the used rate coefficient can be considered an upper limit. Other possible reaction pathways were neglected to avoid additional complexity and uncertainty in the model.

The rate constants for C–H bond scission (reactions 6–8) were automatically generated by EXGAS and were kept unchanged. The rate coefficient for the isomerization (pericyclic rearrangement) of THP to 4-penten-1-ol (C*CCCCOH) (reaction 1) has been calculated at the CBS–QB3 level of theory.

Table 4

Ring corrections to the group additivity method of Benson for thermodynamic properties [52].

Group	$\Delta_f H^\circ$ (298 K) (kcal mol ^{−1})	S° (298 K) (cal mol ^{−1} K ^{−1})
Tetrahydropyran (THP) ring	0.5	18.8
Tetrahydrofuran (THF) ring	5.9	25.3
Cyclohexane ring	0.0	18.8

4.2.2. Hydrogen abstraction

Given the symmetry of THP (Fig. 1) only three types of hydrogen atoms can be abstracted by radicals: at C₂ (or C₆) producing the THP-YL-2 radical, at C₃ (or C₅) producing the THP-YL-3 radical, and at C₄ producing the THP-YL-4 radical. The rate constants for H-abstractions by H, CH₃, and allyl (C₃H₅Y) radicals (reactions 15–17, 24–26, 33–35) were obtained from quantum calculations, whereas rate coefficients for H-abstractions by O (reactions 12–14), OH (reactions 18–20) and OOH (reactions 21–23) were deduced from the Evans–Polanyi correlation proposed by Dean and Bozelli [54]. The rate constants for H-abstractions by other radicals were directly provided by the EXGAS software.

4.2.3. Decomposition of cyclic C₅H₉O radicals

The decomposition of the THP-YL-2, THP-YL-3, and THP-YL-4 radicals is an essential part of the high-temperature chemistry of THP. This set of reactions includes ring-openings (reactions 48–52) and the C–H bond β-scissions (reactions 53–56). Rate constants were obtained from theoretical calculations, apart from reaction 53 which is based on an analogy with THF [51] because the CBS–QB3 calculation failed to locate the transition state for this barrierless reaction.

The ring in the THP-YL-2 radical can open by β -scission of C–O and C–C bonds to form R52C₅H₉OA (reaction 48) and R53C₅H₉OEZ (reaction 49) radicals, respectively. In the case of the THP-YL-3 radical, ring-opening by β -scission leads to the formation of a carbon-centered radical (R50C₅H₉OEZ; reaction 50) or an oxygen-centered radical (R51C₅H₉OZ; reaction 51). It can be noticed that the rate coefficient for C–O bond scission is found to be significantly larger than those for C–C fissions, hence ring-opening via C–O bond fission is favored. In the case of the THP-YL-4 radical, ring-opening leads to the formation of a single product, the R54C₅H₉OEZ radical (reaction 52). The C–H β -scission of THP-YL-2, THP-YL-3, and THP-YL-4 radicals forms dihydropyrans, i.e. 3,4-dihydropyran (3,4-DHP; reactions 53, 54) and 3,6-dihydropyran (3,6-DHP; reactions 55, 56).

4.2.4. Decomposition of acyclic C₅H₉O radicals

The R50C₅H₉OEZ, R51C₅H₉OZ, R52C₅H₉OA, R53C₅H₉OEZ, and R54C₅H₉OEZ radicals, produced directly from the ring-opening of fuel radicals as described above, can decompose by β -scission of C–C and C–O bonds to give smaller (unsaturated) species (reactions 57–61), including C₂H₄ and HCHO, two key products in THP combustion as observed by Labbe et al. [10] and Dagaut et al. [9]. This set of reactions, i.e. β -scissions, has been calculated using the CBS-QB3 method. Rate constants for isomerization by intramolecular H atom abstraction (reactions 83–92) and oxidization (reactions 80, 81) and subsequent decomposition by β -scission (reactions 62–77) were provided by EXGAS.

4.3. Kinetic parameters in the secondary mechanism of THP

The reactions involved in the *primary mechanism* described above lead to formation of a large number of products such as 1-C₄H₈, HCHO, C₂H₄, C₃H₆, 1,3-C₄H₆, CH₃CHO, C₂H₃CHO, C₂H₅CHO, C₄H₆OAZ (which are already included in the C₀–C₆ reaction base), and C₃H₆OEZ, C₄H₆OEZ2, C₅H₈OEZ2, CCCCCHO, C*CCCOC, CCCOC*C, C*CCCCOH, 3,4-DHP, 3,6-DHP (which are not included in the reaction base). The *secondary mechanism* includes the reactions of these primary products. The reactions of the products included in the reaction database will not be discussed here. The reactions of other species of the secondary mechanism are listed in Table S3 (reactions 94–354), and their kinetic parameters are mostly provided by EXGAS. Some rate coefficients were subsequently updated with results from theoretical calculations or by using analogy with structurally similar species. More specifically, retro-Diels–Alder reactions for 3,4-DHP and 3,6-DHP (reactions 263 and 315, respectively) were calculated using the CBS-QB3 method and added in the mechanism. The unimolecular initiations of *n*-pentanal (CCCCCHO) were estimated from *n*-butanal [51,55,56], and those of 4-penten-1-ol (C*CCCCOH) were calculated by the CBS-QB3 method.

4.4. Thermodynamic and transport data

Thermochemical data directly provided and stored as 14 polynomial coefficients by EXGAS (coupled with THERGAS [57] based on the group additivity method proposed by Benson [58]) was used to ensure consistency with rate constants of a large part of reactions generated by this software. This thermochemical data was used also to determine rate constants by Evans–Polanyi correlations. The enthalpy of formation for THP used in the present model agrees with those found in the literature [9,10,59–61] to within 1 kcal mol^{–1}. Further information on the thermochemical parameters for THP is provided in the Supplemental Material (Table S5 and Fig. S3). The transport properties of species, for which no data is available from literature, were estimated by an in-house code that is part of the new EXGAS software package. This code is based on the correlations proposed by Wang and Frenklach [62]. The mechanism, thermodynamic and

transport data are available in CHEMKIN format in the Supplemental Material.

5. Results

5.1. Thermal decomposition in tubular flow reactor

The mole fraction profiles of selected species, measured as a function of the temperature, are presented in Figs. 2–5 for N₂ dilution 90% and residence time 0.5 s, and, additionally, in Figs. S4–S7 (Supplemental Material) for N₂ dilution 96% and residence time 0.2 s. In all these figures, the symbols correspond to experimental results and the lines to simulated results. Simulations without the unimolecular initiations involving C–C and C–O breaking (reactions 1–5 of Table 3) were also performed and correspond to *thin* solid lines in these figures, which will be discussed in Section 6.1.

Figure 2 shows that THP starts to decompose from 920 K and is then consumed rapidly with increasing temperature. The fuel conversion reaches 50% at around 1030 K, and is almost completed at 1120 K. The decomposition of THP leads to the formation of more than twenty products. Light products such as methane (CH₄), carbon monoxide (CO), ethylene (C₂H₄), ethane (C₂H₆), acetylene (C₂H₂), and hydrogen (H₂) were detected in large amounts with mole fractions increasing with temperature. C₂H₄ is the most abundant product in the THP pyrolysis. Its yield increases monotonically, with a maximum mole fraction of $\sim 92 \times 10^{-3}$ at the highest conversion. CO was also detected in high concentrations; its largest mole fraction, $\sim 70 \times 10^{-3}$, is measured at the highest conversion. At the highest temperature, about 75% of oxygen present in the THP molecule is converted to CO. The proposed model predicts the fuel conversion and the formation of these C₀–C₂ products (apart for H₂O) quite well, especially the most abundant ones, concerning both the positions and shapes of the mole fraction profiles as well as their maximum values. In the case of water, a discrepancy between prediction and experiment is observed. This might be due to uncertainties in the measurements because the TCD water signal was quite low, especially at pyrolysis conditions.

The mole fractions of several C₃–C₅ products including C₃H₄ isomers (allene aC₃H₄ and propyne pC₃H₄), propene (C₃H₆), propane (C₃H₈), 1,3-butadiene (1,3-C₄H₆), 1-butene (1-C₄H₈), and 1-pentene (1-C₅H₁₀), are presented in Fig. 3. Among these species, 1,3-C₄H₆ and C₃H₆ are detected with large mole fractions of 80×10^{-4} and 33×10^{-4} , respectively, at 780–800 K. The yields of most of C₃–C₅ species go through maxima at an intermediate temperature (except for C₃H₄ isomers). The two C₃H₄ isomers are well separated with the GC setup used in this study. The present model is in qualitative agreement with all species displayed, i.e. it accurately predicts the temperatures at which the mole fraction profiles increase, decrease or peak. For most species, the calculated mole fraction profiles agree within the stated uncertainty with the experimental results. Note that at higher temperatures, the role of secondary, molecular growth, chemistry becomes increasingly important, evidenced by the increase in experimental aromatic yield (Fig. 4). The corresponding reactions are part of the base mechanism and not the focus of this work. These reactions can however have an impact on the model performance at the highest investigated temperatures.

Figure 4 shows the evolution of the mole fraction of C₅–C₁₀ unsaturated cyclic products: 1,3-cyclopentadiene (C₅H₆), and aromatic species such as benzene (C₆H₆), toluene (C₇H₈), styrene (C₈H₈), indene (C₉H₈), and naphthalene (C₁₀H₈). These species are important soot precursors, and their profiles exhibit the typical exponential increase as a function of temperature. Among these cyclic products, C₅H₆ and C₆H₆ are the most abundant ones with a mole fraction of $\sim 15 \times 10^{-4}$ and $\sim 30 \times 10^{-4}$, respectively, at the highest temperature. Other species were detected at lower levels, less than 4×10^{-4} at the highest temperature. Overall, the present kinetic model is able to predict the formation of these important soot precursors well. In

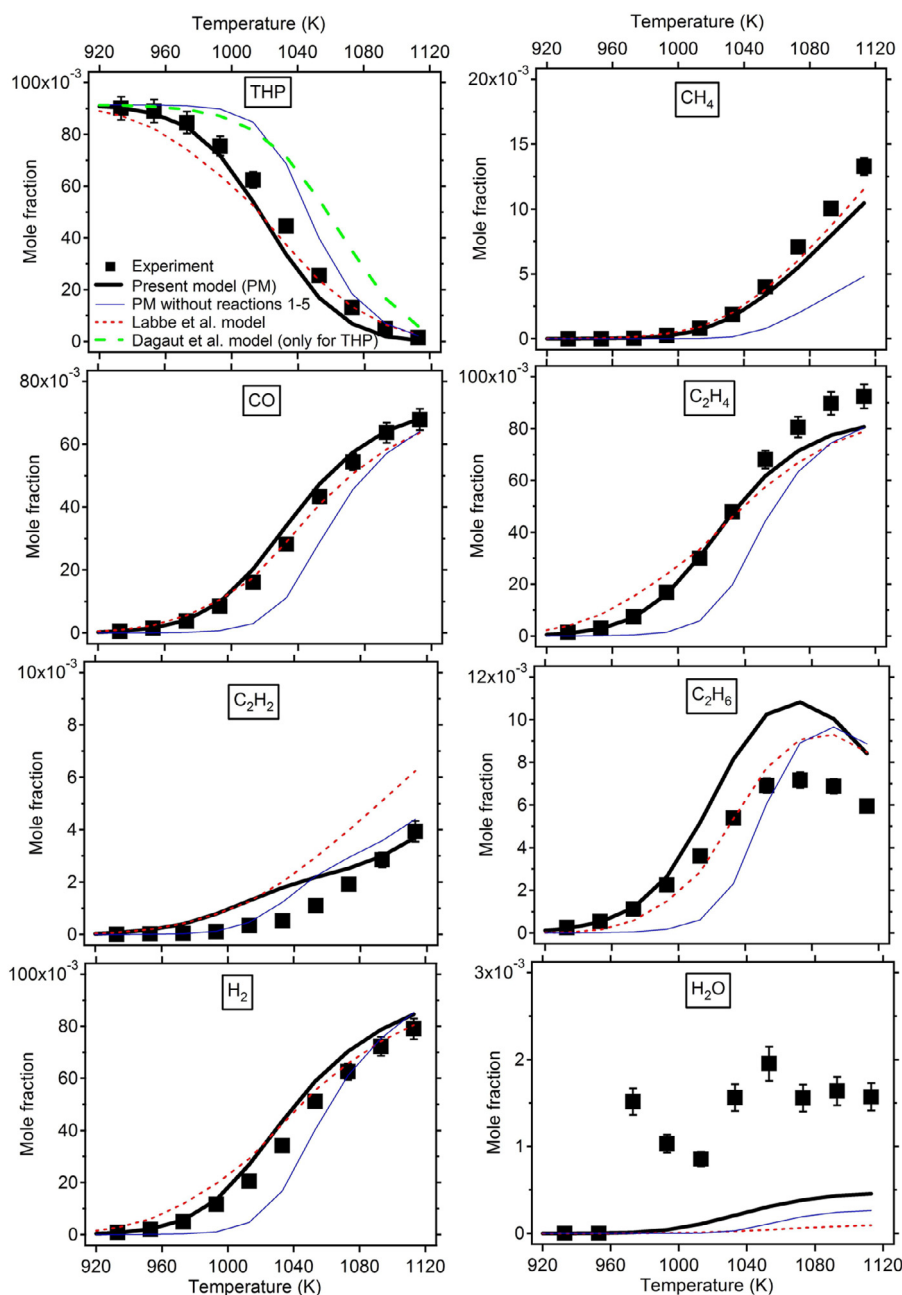


Fig. 2. Evolution of the mole fraction of fuel and C_0 – C_2 products of THP pyrolysis at a residence time of 0.5 s, 90% N_2 dilution, and a pressure of 0.17 MPa. Symbols: experiments; lines: simulations. Mole fraction profiles for higher dilution (96%) can be found in the Supplemental Material.

the case of naphthalene, a different rate coefficient for the recombination of two cyclopentadienyl radicals was also considered. Figure 4 shows that the use of the values proposed by Slavinskaya and Frank [63] and Djokic et al. [15] results in an overprediction of the formation of naphthalene. A significant underprediction is found when the kinetic parameters recently proposed by Slavinskaya and co-workers [64] were used. This suggests that further studies are needed to improve the accuracy of the rate constants for cyclopentadienyl radical recombination.

Figure 5 presents the mole fraction profiles of oxygenated intermediates including acrolein (C_2H_3CHO) and formaldehyde ($HCHO$), as a function of temperature. These two species were detected in high amounts; the peak mole fractions are $\sim 1.2 \times 10^{-3}$ and $\sim 7 \times 10^{-3}$ for C_2H_3CHO and $HCHO$, respectively. A quite good agreement between

experimental and simulation results is observed for these two toxic species.

As mentioned earlier, an additional measurement of the THP pyrolysis at higher N_2 dilution (96%) and shorter residence time (0.2s) was also conducted, and the results are presented in Figs. S4–S7 (Supplemental Material). Overall, the agreement between experiments and simulations is comparable to that observed at lower dilution conditions.

To provide a comparison between the present model and those by Labbe et al. [10] and Dagaut et al. [9], the pyrolysis data has been simulated with the latter two. The predictions with the Labbe et al. model are included in Figs. 2–5 in the main text and in Figs. S4–S7 in the Supplemental Material. Simulation results obtained with the Dagaut et al. model are presented in Figs. 2 and S4 (only THP)

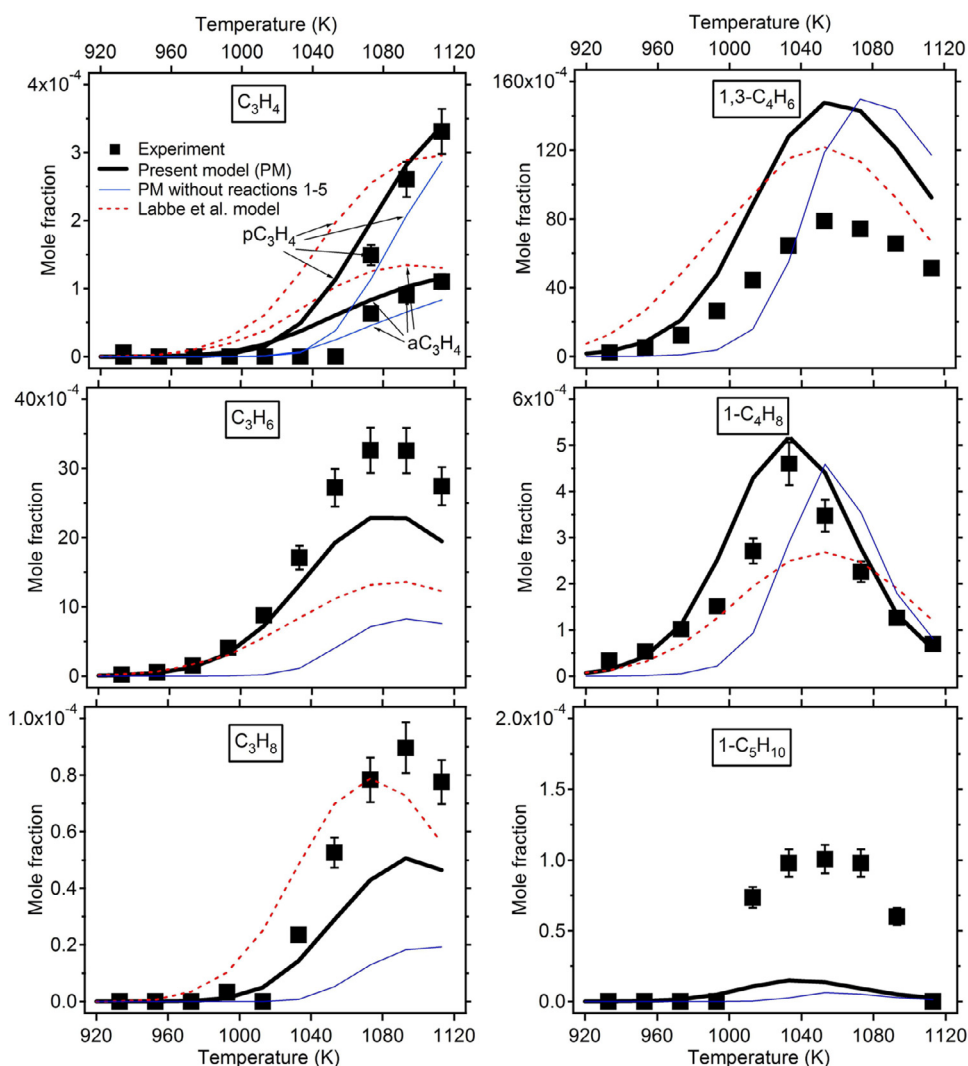


Fig. 3. Evolution of the mole fraction of C_3 – C_5 products of the THP pyrolysis at a residence time of 0.5 s, 90% N_2 dilution, and a pressure of 0.17 MPa. Symbols: experiments; lines: simulations. Mole fraction profiles for higher dilution (96%) can be found in the Supplemental Material.

and in Figs. S8–S13 (other species). The Labbe et al. model predicts the fuel conversion and mole fraction profiles of products well and its overall performance is similar to the present model. Note that the model by Labbe et al. does not contain reactions for the formation of C_{6+} aromatics, while the present model includes these heavy species and predicts quite well their formation. The Dagaut et al. model underestimates the overall reactivity of THP at the pyrolysis conditions studied. One important difference between the new and the previous mechanisms is the description of the initial unimolecular ring-opening reactions. Dagaut et al. and Labbe et al. adopted the reaction steps proposed by Lifshitz et al. [65] based on their shock tube experiments on THF. According to Lifshitz et al., the diradicals formed in the ring opening bond scission step dominantly undergo subsequent β -scission reactions to create smaller species. In contrast, the current model contains diradical chemistry which favors intramolecular H transfer over β -scissions. The preference for H migration is supported by earlier studies [51,66,67]. Since the formation of diradicals is the rate determining step, differences in the branching ratios have no significant impact on the overall reactivity (both channels yield solely stable product). Furthermore, once the radical pool is established, the product distribution is mainly controlled by abstraction reactions and not by the initial bond scission reactions.

5.2. Species profiles in low-pressure premixed flat flame

In the present study, the chemical structure of the THP flames was investigated for two equivalence ratios, i.e. $\varphi = 1.0$ and 1.3. These flames are leaner than that studied by Labbe et al. [10]. A large number of intermediates and products were identified and quantified using on-line GC-TCD-FID/methanizer and GC-MS analyses. Note that Labbe et al. used MBMS to investigate the laminar premixed low-pressure THP flame. The analysis techniques used in both studies are hence complementary which allows an independent confirmation of the species identified in both studies. The choice of different equivalence ratios provides additional data for kinetic model validation.

The flame temperature profiles and the mole fraction profiles of selected species are presented and discussed in this section. Figures 6–12 present the temperature and mole fraction profiles of C_0 – C_6 species (reactant, major and intermediate products) as a function of the distance above the burner (h). These figures show that the reaction zone peaks at ~ 1 – 3 mm above the burner. Simulations without the unimolecular initiations involving C–C and C–O scission (reactions 1–5 of Table 3) were also performed and the results are included in the corresponding figures (thin solid lines), but *only* for species, whose profiles were affected by the absence of these reactions. The differences will be discussed in Section 6.2.

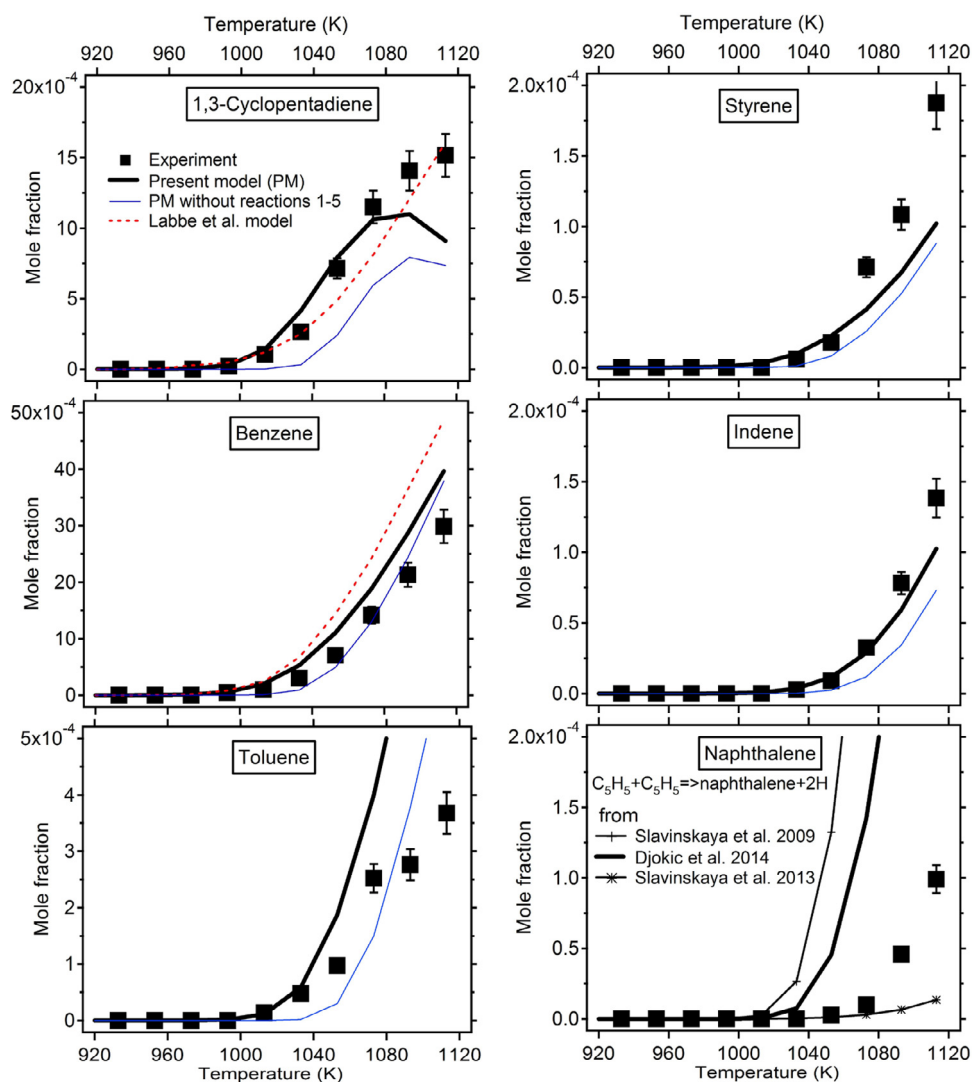


Fig. 4. Evolution of the mole fraction of cyclic products of the THP pyrolysis at a residence time of 0.5 s, 90% N_2 dilution, and a pressure of 0.17 MPa. Symbols: experiments; lines: simulations. Mole fraction profiles for higher dilution (96%) can be found in the Supplemental Material.

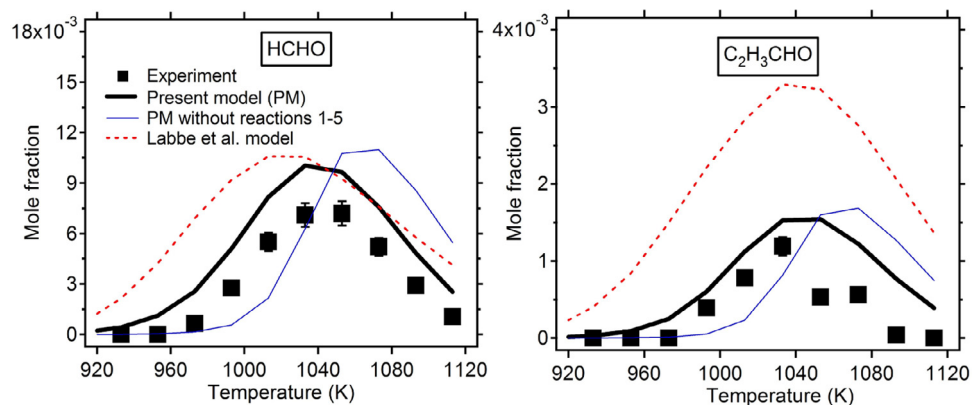


Fig. 5. Evolution of the mole fraction of oxygenated intermediates of the THP pyrolysis at a residence time of 0.5 s, 90% N_2 dilution, and a pressure of 0.17 MPa. Symbols: experiments; lines: simulations. Mole fraction profiles for higher dilution (96%) can be found in the Supplemental Material.

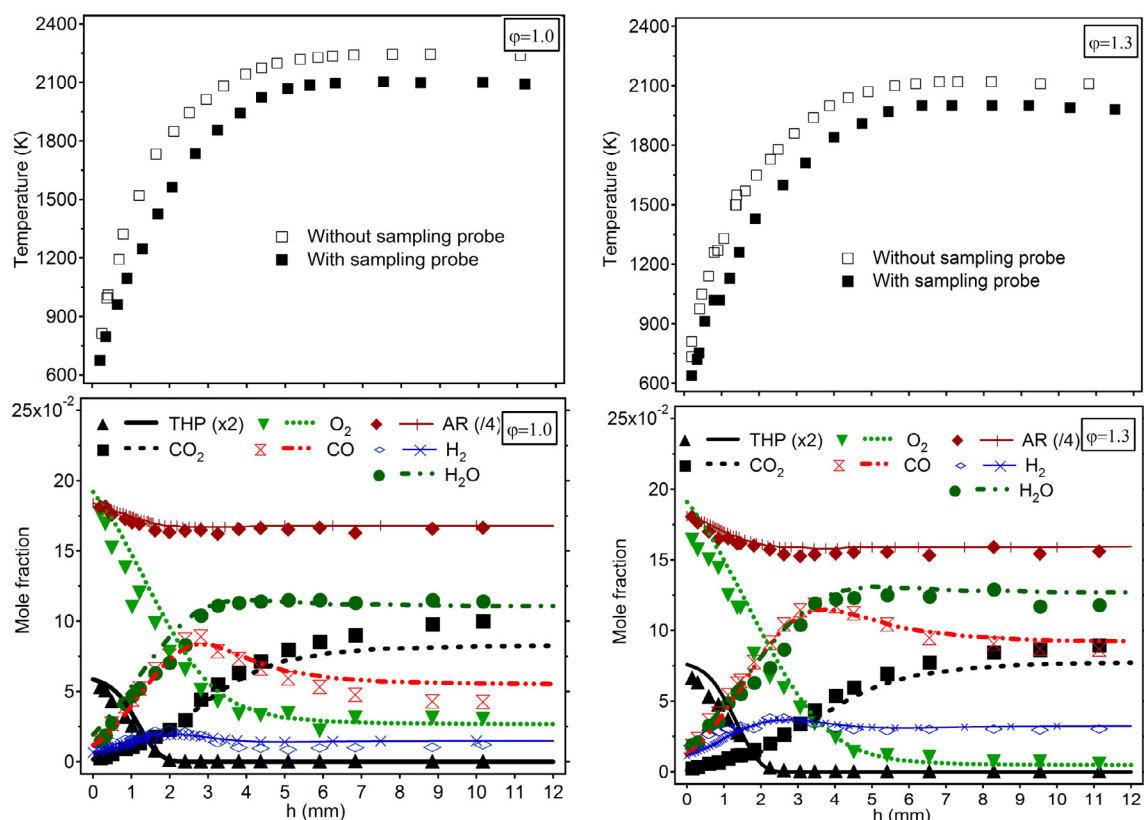


Fig. 6. Flame temperature and mole fraction profiles of major species (fuel, O_2 , Ar, CO_2 , CO, H_2O , and H_2) as a function of height above burner (h) for two THP flames ($\phi = 1.0$ and 1.3). Symbols: experiments; lines: simulations with the present model.

Figure 6 displays the temperature profiles measured for the two flames ($\phi = 1.0$ and 1.3) with and without the sampling probe, as well as the mole fraction profiles of reactants (THP and O_2), diluent (Ar), and main products (CO , CO_2 , H_2O , and H_2). The presence of the sampling probe induces a thermal perturbation, therefore, temperatures measured with the probe are lower than those measured without the probe. In latter case, the maximum values were around 2100–2200 K (at $h > 6$ mm) (Fig. 6). THP is completely consumed at $h > 2.0$ mm, while a significant mole fraction of O_2 remains in the post flame region. As expected, the mole fraction of CO_2 formed in the stoichiometric flame is larger than that in the fuel-rich flame, with the opposite trend found for CO and H_2 . The profiles of CO display marked maxima (~ 9 – 12×10^{-2}) at a height of 2.5–3.5 mm. CO is present in high amounts in the post flame region, even in the stoichiometric flame. The evolution of the CO mole fraction as a function of fuel conversion in the THP flame (linear fit from 15 to 90% fuel conversion: $X_{\text{CO(THP)}} = 7 \times 10^{-4} Y_f + 0.0059$; where X_{CO} is CO mole fraction and Y_f is fuel conversion) closely resembles that measured in a THF flame ($X_{\text{CO(THF)}} = 7 \times 10^{-4} Y_f + 0.0053$) [11] under similar conditions. This implies that the CO elimination process is similar for THP and THF. The present model is able to reproduce the consumption of reactants (THP and O_2), the formation of main products (CO_2 , CO, H_2O , and H_2), and the diluent (Ar) profile, both with regard to the profile shape and the mole fraction values.

Figure 7 displays the mole fraction profiles of C_1 – C_2 hydrocarbon intermediates, including methane (CH_4), acetylene (C_2H_2), ethylene (C_2H_4), and ethane (C_2H_6). C_2H_4 is the most abundant intermediate in the THP flames, with maximum mole fractions of $\sim 12 \times 10^{-3}$ and $\sim 16 \times 10^{-3}$ in the stoichiometric and fuel-rich flames, respectively. This is consistent with results obtained in pyrolysis, and in flames of THF [11], while the combustion of *unsaturated* cyclic ethers, such as furanic fuels, produced acetylene as the most abundant intermedi-

ate species [22,68–70]. This is in agreement with previous observations on the formation of these two major intermediates in the combustion of *saturated* and *unsaturated* fuels by Togbé et al. [71]. In the THP flames, the C_1 and C_2 intermediates are present in equivalent amounts to those obtained in THF flame [11]. Figure 7 shows that the mole fraction profiles of all the C_1 – C_2 hydrocarbon intermediates are very well predicted by the present model for both equivalence ratios.

Figure 8 presents the mole fraction profiles of several C_3 hydrocarbon intermediates, which are the two C_3H_4 isomers propyne (pC_3H_4) and allene (aC_3H_4), propene (C_3H_6), and propane (C_3H_8). C_3H_6 is the most abundant C_3 intermediate, with a maximum mole fraction of approximately 11×10^{-4} and 13×10^{-4} in the stoichiometric and fuel-rich flames, respectively. C_3H_8 is the second most abundant product among the C_3 hydrocarbon species (peak mole fractions $\sim 5 \times 10^{-4}$ – 6×10^{-4}). Similarly to results obtained for C_1 and C_2 species, most of the C_3 compounds are produced in similar amounts in the THP and THF [11] flames. The formation of these C_3 species is well reproduced by the present model except for aC_3H_4 , for which concentration is over-predicted by more than a factor of two.

Figure 9 shows the mole fraction profiles of C_4 hydrocarbon intermediates in the two THP flames. The data includes 1,3-butadiyne (C_4H_2), 1,3-butadiene ($1,3\text{-C}_4\text{H}_6$), the two C_4H_8 isomers 1-butene ($1\text{-C}_4\text{H}_8$) and 2-butene ($2\text{-C}_4\text{H}_8$) ($1\text{-C}_4\text{H}_8$), and n -butane ($n\text{-C}_4\text{H}_{10}$). 1,3- C_4H_6 is the most abundant C_4 species with maximum mole fractions of $\sim 8 \times 10^{-4}$ (at $\phi = 1.0$) and $\sim 13 \times 10^{-4}$ (at $\phi = 1.3$). Note that with the GC setup used in this study C_4H_4 cannot be separated from 1,3- C_4H_6 . The study of a rich THP flame by Labbe et al. [10] revealed that C_4H_4 is formed at an about ten times lower level than 1,3- C_4H_6 . Therefore the GC signal in the current study was exclusively assigned to 1,3- C_4H_6 .

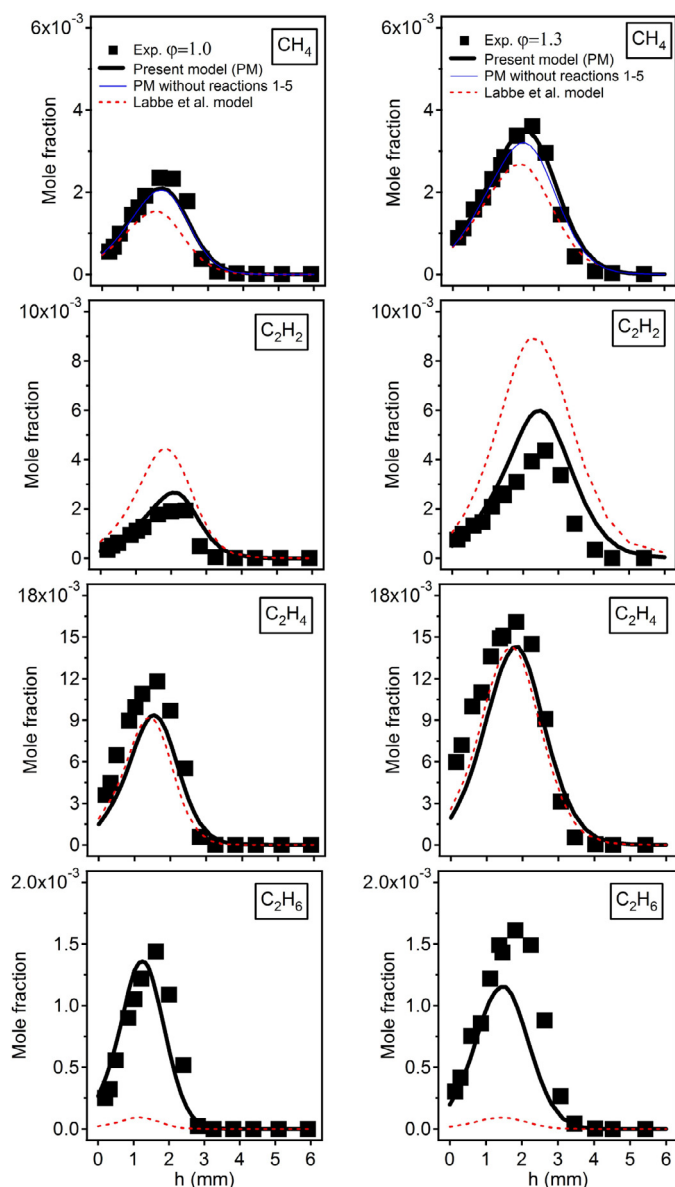


Fig. 7. Mole fraction profiles of C_1 – C_2 hydrocarbon intermediates in two THP flames ($\phi = 1.0$ and 1.3). Symbols: experiments; lines: simulations.

The 1,3- C_4H_6 mole fractions measured in the present THP flames are much higher than that (1.2×10^{-4} at $\phi = 1.3$) measured in the THF flame [11] under similar flame conditions and this trend is also observed for C_{4+} intermediates. In comparison to 1,3- C_4H_6 , C_4H_8 isomers and nC_4H_{10} were produced in smaller amounts. C_4H_2 is present in very low quantities, i.e. less than 10 ppm. In general, the model predicts the formation of these C_4 species well, especially the abundant ones. C_4H_2 is an exception as its formation in the two THP flames of this study is severely over-predicted. However, the opposite observation is made when simulating the experimental THP data of Labbe et al. [10] (see in the Supplemental Material).

Figure 10 presents profiles of the C_5 – C_6 hydrocarbon intermediates, including 1,3-cyclopentadiene (C_5H_6), two C_5H_{10} isomers (1-pentene and 2-pentene), and benzene (C_6H_6). All species were formed in small amounts (<50 ppm). In particular cyclic unsaturated species, C_5H_6 and C_6H_6 , known as important soot precursors, were detected in very low mole fraction (less than 13 ppm). A similar trend was observed in the combustion of THF by Tran et al. [11] at comparable conditions. The model predicts the mole fraction profiles of

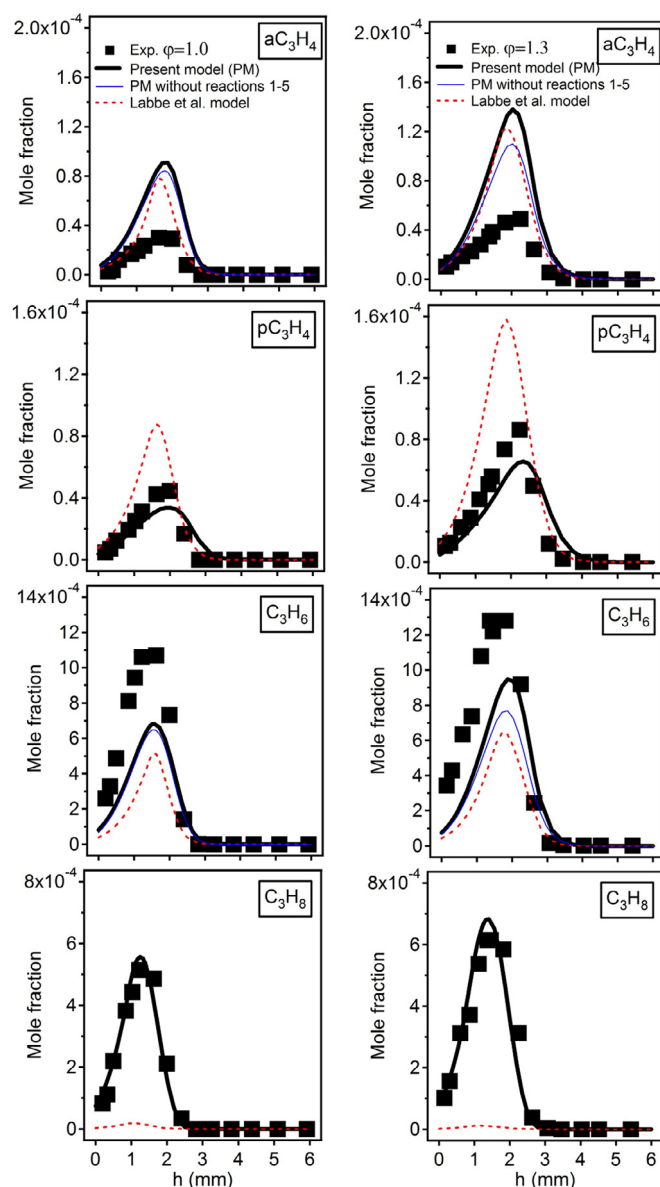


Fig. 8. Mole fraction profiles of C_3 hydrocarbon intermediates in two THP flames ($\phi = 1.0$ and 1.3). Symbols: experiments; lines: simulations.

C_5H_6 and C_5H_{10} isomers reasonably well but underpredicts the formation of C_6H_6 . Note that this last species is very well predicted in the pyrolysis experiments (Fig. 4).

A large number of oxygenated intermediates were identified and quantified in the THP flames. They are presented in Figs. 11 and 12. Special attention is paid to toxic products such as formaldehyde ($HCHO$), acetaldehyde (CH_3CHO), and acrolein (C_2H_3CHO). Among the oxygenated intermediates, $HCHO$ is present with the largest mole fractions ($\sim 5 \times 10^{-3}$ in both flames). This value is higher than that measured in THF flame (peak mole fraction $\sim 3 \times 10^{-3}$) [11]. Note that, $HCHO$ is a primary product in the combustion of these oxygenated fuels. CH_3CHO (Fig. 11) and C_2H_3CHO (Fig. 12) are other abundant oxygenated intermediates in the combustion of THP, with peak mole fractions of $\sim 0.7 \times 10^{-3}$ and $\sim 0.75 \times 10^{-3}$, respectively. Other oxygenated intermediates, such as ethylene oxide ($c\text{-}CH_2OCH_2$), dimethylether (DME), ethanol (Fig. 11), propanal, acetone, and 3,4-dihydropyran (3,4-DHP) are present in smaller mole fractions (Fig. 12). The latter species is directly related to the decomposition of

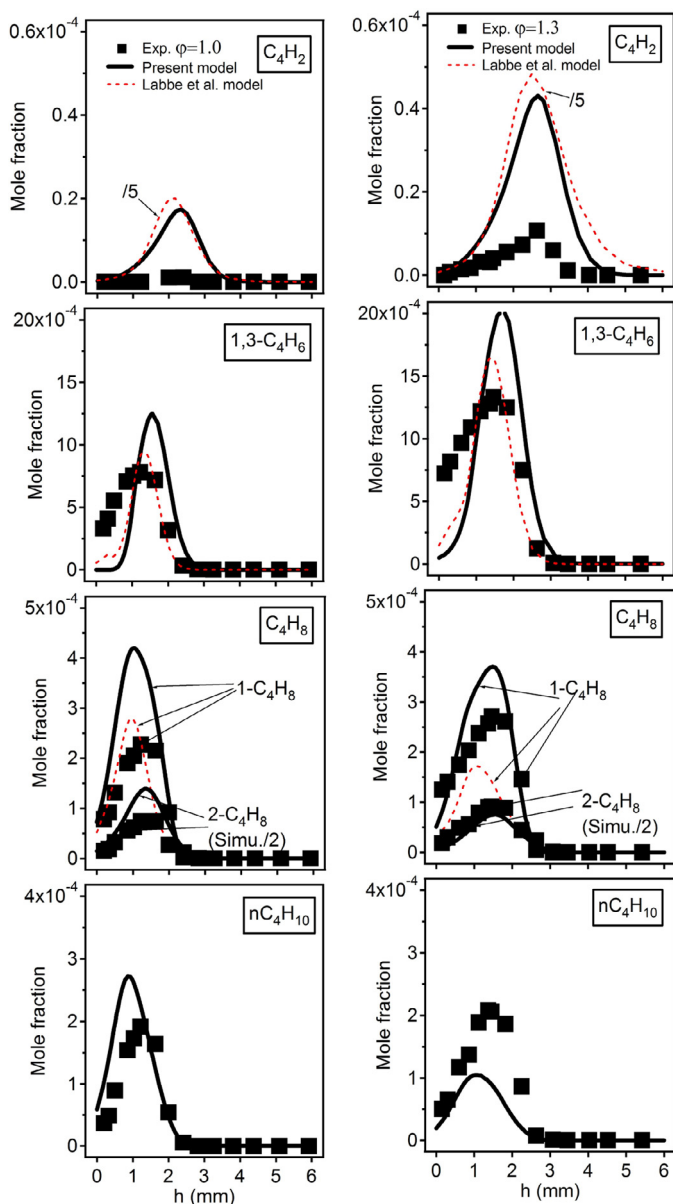


Fig. 9. Mole fraction profiles of C_4 hydrocarbon intermediates in two THP flames ($\varphi = 1.0$ and 1.3). Symbols: experiments; lines: simulations.

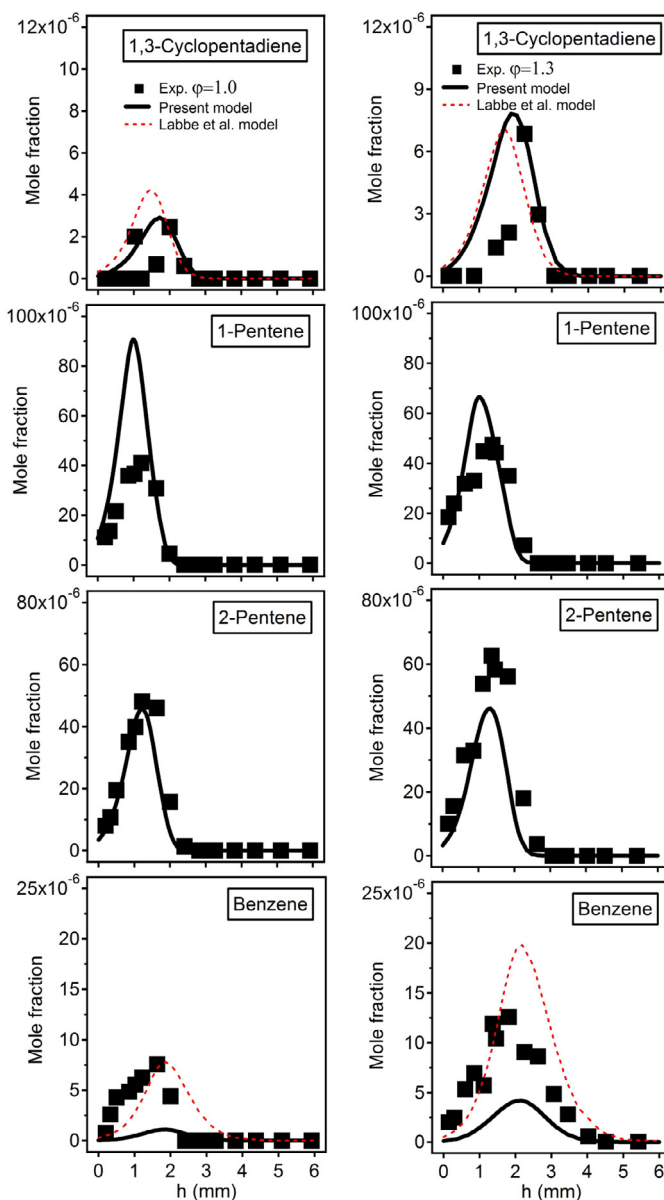


Fig. 10. Mole fraction profiles of C_5 – C_6 hydrocarbon intermediates in two THP flames ($\varphi = 1.0$ and 1.3). Symbols: experiments; lines: simulations.

THP as will be discussed in Section 6. Figures 11 and 12 show good agreements between experimental and simulation results, especially for important species.

Simulations with the model of Labbe et al. have been also included in Figs. 7–12 (dotted lines). In general, this kinetic model predicts the flame structure quite well. Both mechanisms perform similarly well for C_2H_4 , aC_3H_4 , C_5H_6 , $HCHO$, and CH_3CHO . The predictions with Labbe et al.'s model are superior for a few species, e.g. 1,3- C_4H_6 and C_6H_6 . Note that both model used the same reactions subset for C_6H_6 [50], thus differences in C_6H_6 predictions are caused by differences in the base mechanism. The present model yields more accurate predictions for numerous intermediates measured in the studied flame conditions, especially C_2H_2 , C_2H_6 , C_3H_8 , ethylene oxide ($c\text{-}CH_2OCH_2$), and 3,4-DHP. Some species such as $n\text{-}C_4H_{10}$, 2- C_4H_8 , 1-pentene, 2-pentene, DME, and ethanol are not included in the model of Labbe et al., and the present model predicts those well.

In order to test the performance of the present kinetic model with respect to data from another independent experimental setup, the

structure of the fuel-rich THP flame ($\varphi = 1.75$, Ar dilution = 25%, $P = 2.66$ kPa) of Labbe et al. [10] was simulated. Figures S14–S17 in the Supplemental Material show the results of these simulations. It can be seen that the predictions generally agree with the trends in the experimental data of Labbe et al. [10]. However, because of the significant noise in the data, a more stringent validation is not possible.

5.3. Laminar burning velocities at atmospheric pressure

Figure 13 presents the measured and predicted adiabatic burning velocities of THP flames at atmospheric pressure and unburned gas temperatures ranging from 298 to 398 K and equivalence ratios ranging from 0.55 to 1.50. As expected, the laminar burning velocity increases with increasing the initial temperature. The maximum flame burning velocity is found in the slightly rich mixture ($\varphi = 1.1$). For the simulation of laminar burning velocities, the mechanism was reduced in order to achieve convergence. Reactions of some large species and their derived radicals such as 3-butenyl-benzene,

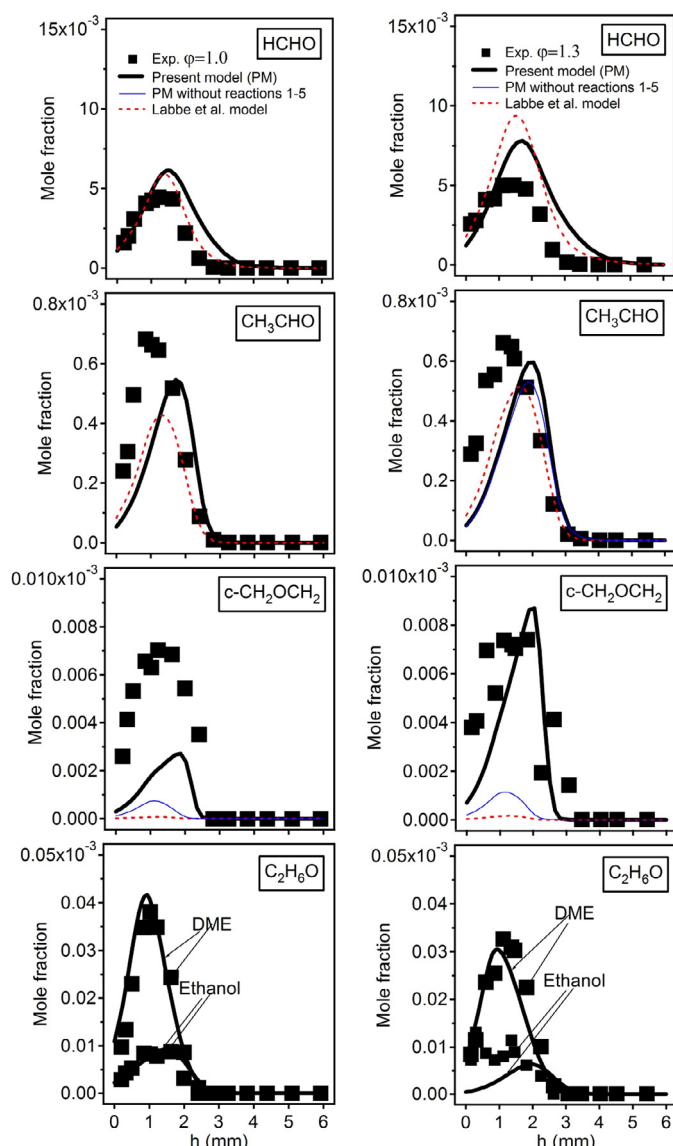


Fig. 11. Mole fraction profiles of C_1 – C_2 oxygenated intermediates in two THP flames ($\phi = 1.0$ and 1.3). Symbols: experiments; lines: simulations.

cumene, bibenzyl, cresol, benzofuran and 3-hydroxy-benzaldehyde were removed. Figure 13 shows that the present model (thick lines) captures the effect of equivalence ratios and initial temperatures well despite the fact that a slight overprediction is found for lean and stoichiometric mixtures.

Comparing between the THP and THF [11] flames, the ether with the larger ring leads to faster flame propagation. Indeed, the maximum adiabatic laminar burning velocity of THP (44 cm s^{-1} , measured at $T = 298 \text{ K}$) is slightly higher than that of THF (43 cm s^{-1} , measured at $T = 298 \text{ K}$) [11], this implies that THP is more reactive than THF. The trends pointed out above were also observed for cyclic hydrocarbon fuels in an earlier comparative study of Davis and Law [72].

5.4. Auto-ignition in shock tube

Figure 14 presents the experimental and modeling results obtained for the ignition delay times (t_{igni}) of THP-oxygen-argon mixtures containing 0.15, 0.25, 0.5, and 1% of fuel for equivalence ratios of 0.5, 1.0 and 2.0. The following trends are observed: (i) For all studied mixtures, the ignition delay times decrease as temperature rises; (ii) For a given fuel concentration (Fig. 14a), the ignition delay time

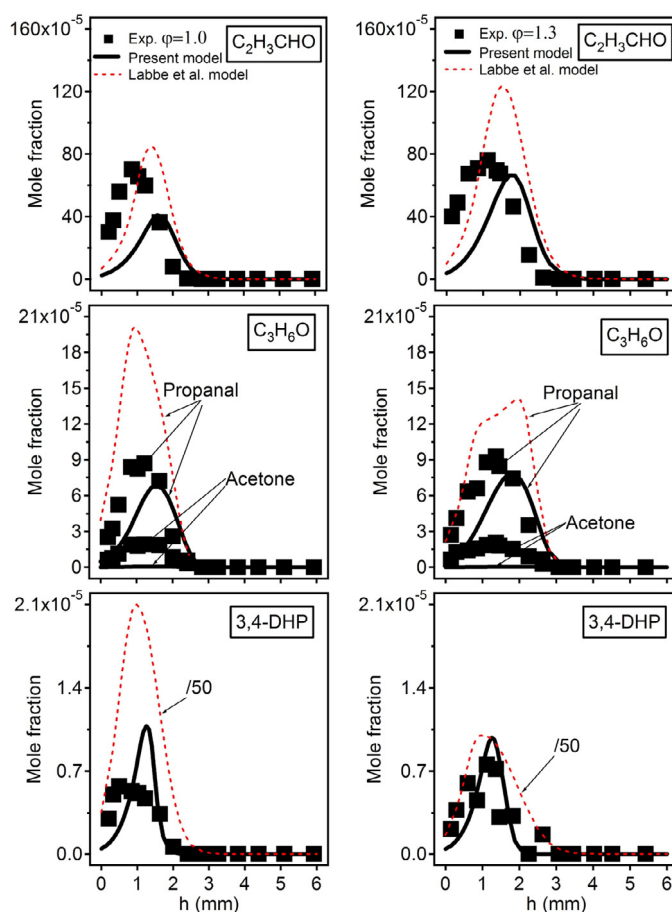


Fig. 12. Mole fraction profiles of C_3 – C_5 oxygenated intermediates in two THP flames ($\phi = 1.0$ and 1.3). Symbols: experiments; lines: simulations.

increases when going from fuel-lean to fuel-rich conditions; (iii) For a given equivalence ratio (Fig. 14b), the ignition delay time increases when the fuel concentration is lowered from 1 to 0.15%.

For fuel/oxygen mixtures, the determination of power dependences is often proposed from the overall statistical correlation between t_{igni} and the gas concentrations shown in Eq. (2):

$$t_{\text{igni}} = A \exp(E/RT) [\text{fuel}]^a [\text{O}_2]^b [\text{Ar}]^c \quad (2)$$

where A is the pre-exponential factor, R the gas constant (in this work, in $\text{cal K}^{-1} \text{ mol}^{-1}$) and E the apparent “activation energy”. For a restricted range of pressure and temperature, a , b and c are usually constants. Such a statistical correlation has been derived from the present experiments, but since the mole fraction of argon had small variations under the different initial conditions (from 92.0 to 98.8%) and seems not to affect the delay times, it was chosen to keep $c = 0$. A multi-linear regression leads to the following relationship, with the concentrations behind the reflected shock wave given in mol cm^{-3} :

$$t_{\text{igni}}(\text{s}) = 1.58 \times 10^{-18} \exp(74,913/RT) [\text{THP}]^{1.12} [\text{O}_2]^{-1.68} \quad (3)$$

Compared to THF, the ignition delay times of THP are slightly shorter at similar conditions (Fig. 14a and b). This is consistent with the slightly higher flame speeds of THP compared to THF as discussed in Section 5.3. Figure 14a and b shows that although the model tends to underpredict the measured slopes, it reproduces well the influences of fuel concentration, equivalence ratio, and temperature on ignition delay times. A comparison between simulations with (thick lines) and without (thin lines) the unimolecular initiations involving C–C and C–O scissions (reactions 1–5 of Table 3) is presented in Fig. 14c and will be discussed in Section 6.3. Ignition delay time

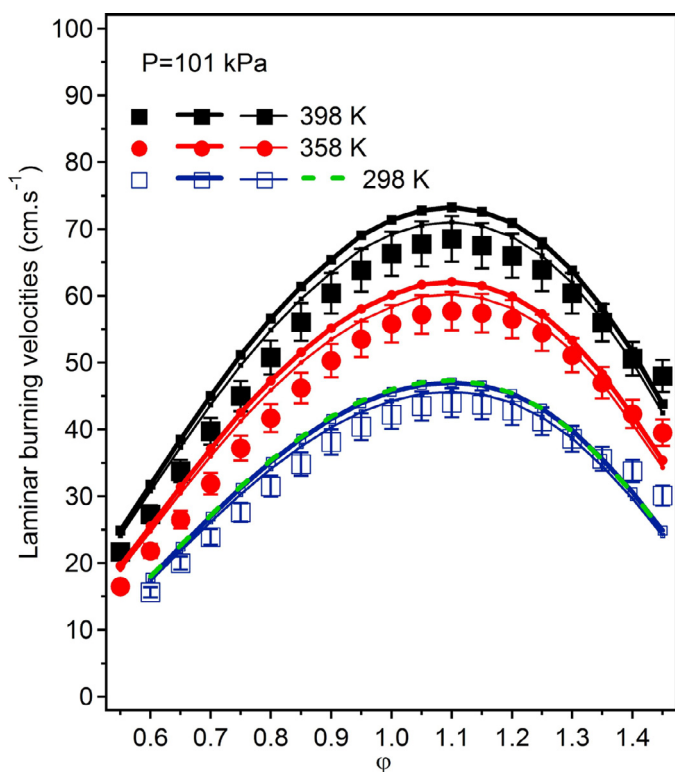


Fig. 13. Laminar burning velocities for THP-air flames at 101 kPa (1 atm). Symbols: experiments; lines: simulations (thick lines: present model; dotted line at 298 K: model without the unimolecular initiations involving C–C and C–O breaking; thin lines: model with rate constant of reaction $\text{O}_2 + \text{H} = \text{OH} + \text{O}$ proposed by Hong et al. [76]).

simulations with the models of Dagaut et al. [9] and Labbe et al. [10] have been also included in Fig. 14d and Fig. S18 (Supplemental Material). The predictions by the models of Labbe et al. and this study are in close agreement, and much different to those by the pioneering model of Dagaut et al. This comparison helps to view the overall performance of these three kinetic models under given ignition situations. A full comparative sensitivity study between these models is beyond the main focus of the present work, however.

All experimental data obtained in the present study are available as tabulated results in the Supplemental Material.

6. Model analysis

To gain more insight in the pyrolysis and combustion chemistry of THP, rate and sensitivity analyses were performed using CHEMKIN. The main reaction pathways of THP consumption for pyrolysis conditions and flame experiments are shown in Fig. 15 and Fig. 16, respectively. The structure and nomenclature of selected species relevant to the following discussion in this section are shown in Table S4 (Supplemental Material).

6.1. Reaction pathways of THP thermal decomposition

Figure 15 presents the simulated rate-of-production analysis for THP consumption under the pyrolysis conditions for a temperature of 1053 K, at 500 mm from the reactor inlet corresponding to ~50% of fuel conversion, and at two earlier stages in the plug flow reactor (at 100 mm and 160 mm; with lower fuel conversions around 0.1–0.5%), using the developed kinetic model. A rate-of-production analysis using the Labbe et al. model was also performed and can be found in Supplemental Material (Fig. S19). The main differences between both kinetic models are related to the differences in THP unimolecular decomposition chemistry, as discussed earlier. In the beginning of the

reactor, unimolecular initiations play an important role in fuel consumption, as shown in Fig. 15. At later stages such as from 500 mm (~50% fuel conversion), fuel is mostly consumed by H-abstractions by small *non-oxygenated* radicals (H , CH_3 , C_2H_3 , and C_2H_5) which become abundant. Similar trends can be observed with the Labbe et al. model (Fig. S19).

From the unimolecular initiations two dominant reactions are ring-opening via O–C₂ bond scission (or O–C₆) and C₂–C₃ bond scission (or C₅–C₆) yielding the diradicals B15 and B16 (Fig. 15). The pericyclic rearrangement producing an unsaturated alcohol and ring-opening by C₃–C₄ (or C₄–C₅) bond scission contributes to the fuel consumption at low conversion. The initial products of the ring-opening reactions subsequently decompose and form smaller species.

Hydrogen abstraction from the C₂ (or C₆) position of THP (the carbon atoms bound to the oxygen atom), which produces the THP-YL-2 radical, is the most important hydrogen-abstracting channel, followed by H-abstractions from the C₃ (or C₅) position and those from the C₄ position, yielding the THP-YL-3 and THP-YL-4 radicals respectively. The contributions of these three H-abstraction channels with respect to fuel consumption are consistent with the difference in the dissociation energies of C–H bonds and number of H-atoms between the different positions in the THP molecule. As shown in Fig. 1, the bond dissociation energy (BDE) of a C₂–H bond (95.0 kcal mol^{−1}) is much lower than those for a C₃–H bond (98.7 kcal mol^{−1}) and a C₄–H bond (98.6 kcal mol^{−1}).

As mentioned earlier, the contribution of H-abstractions to the fuel conversion increases along the length of the plug flow reactor, and from 500 mm most of THP is consumed by H-abstractions. This implies that radical reactions dominate the pyrolysis chemistry of THP in line with other oxygenated fuels [73] and hydrocarbons [74]. Therefore, in order to clarify the contributions of the unimolecular initiation reactions involving C–C and C–O scissions to the THP pyrolysis chemistry, additional simulations were performed using the present model without the reactions 1–5 of Table 3, see Figs. 2–5. It is seen that the unimolecular initiations involving C–C and C–O scissions clearly play a role in the THP pyrolysis. Indeed, without these reactions the model predicts the pyrolysis data such as fuel conversion and some product profiles unsatisfactorily. The demonstrated importance of the unimolecular initiations, combined with the fact that the current model uses rate coefficients with considerable uncertainty for these reactions, suggests that further theoretical studies are needed in the near future.

6.2. Reaction pathways of THP combustion

The consumption of THP (Fig. 16) in the stoichiometric low-pressure flame has been analyzed at $h \sim 1.35$ mm, corresponding to a temperature of 1053 K (the same temperature as that used for the THP pyrolysis pathway analysis above) and 70% fuel conversion. Under these conditions hydrogen abstractions are responsible for approximately 99% of the total THP consumption. Hydrogen abstractions mainly occur at the C₂ positions of THP (the carbon atoms bound to the oxygen atom, see Fig. 1) by flame-propagating radicals, such as H , OH , O , and CH_3 . H-abstractions from the C₃ (or C₅) and C₄ position account for approximately 18% and 11% of total THP consumption. The subsequent reactions of these three fuel radicals (THP-YL-2, THP-YL-3, THP-YL-4) play an important role in the overall kinetic mechanism of THP. Therefore, good rate coefficients for these reactions are needed, and hence an accurate method such as CBS-QB3 has been used for the calculations of these rates.

The reactions going via unimolecular initiation involving C–C and C–O bond scissions (reactions 1–5 of Table 3) have a significant impact under the pyrolysis conditions as mentioned in the previous section, whereas these reactions play only a very minor role in the fuel consumption and in the formation of products under the flame

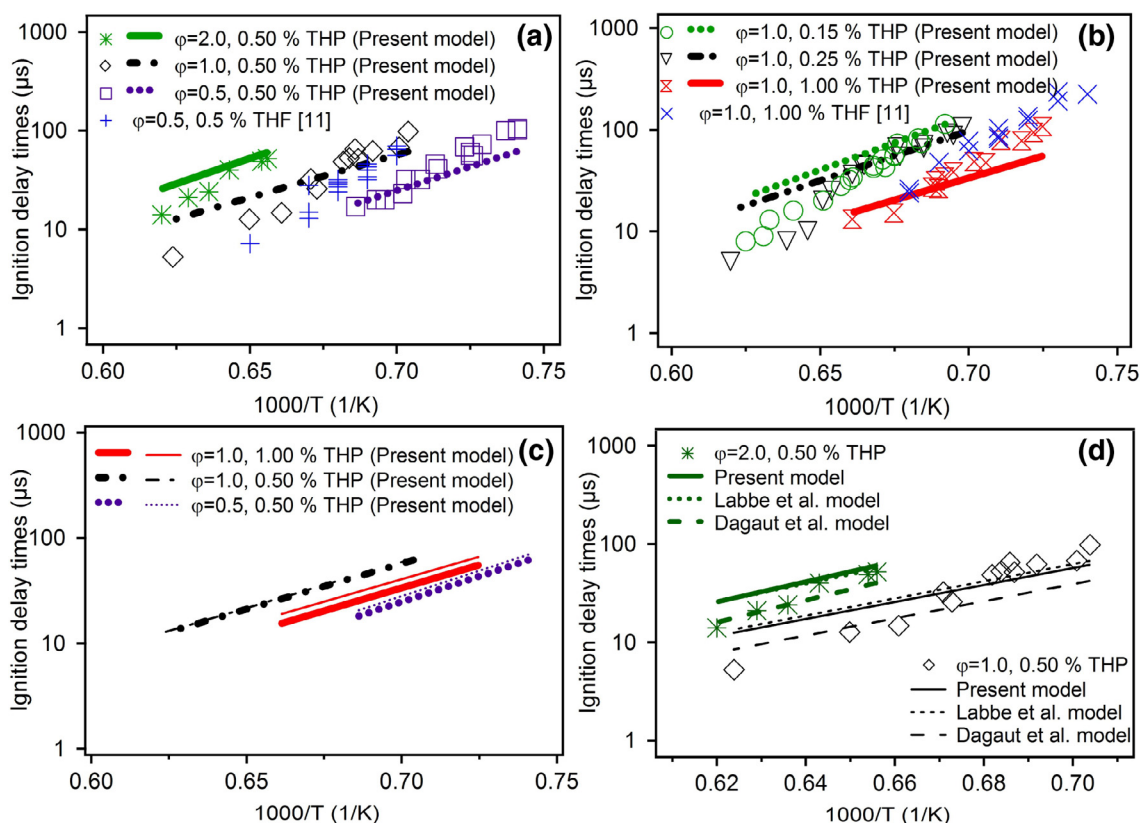


Fig. 14. Ignition delay times of THP in a shock tube: (a) for an initial mole fraction of 0.5% for $\phi = 0.5, 1.0$ and 2.0 ; (b) for $\phi = 1.0$ for initial mole fractions of 0.15 up to 1% ; (c) comparison between simulations with and without the reactions 1–5 of Table 3; (d) comparison to Dagaut et al. [9] and Labbe et al. [10] models. Data obtained for THF [11] at $\phi = 0.5$ (0.5% THF/ 5.5% O_2 / 94% Ar) and $\phi = 1.0$ (1.0% THF/ 5.5% O_2 / 93.5% Ar) are added for comparison. Symbols: experiments; lines: simulations. Thin lines in (c): simulations without the reactions 1–5.

conditions. Simulations without these reactions for the flame structures show only a very minor decrease of the CH_4 (Fig. 7), aC_3H_4 , C_3H_6 (Fig. 8), CH_3CHO (Fig. 11) profiles, and a somewhat stronger decrease of the oxirane ($c\text{-}CH_2OCH_2$) concentration (Fig. 11). Note that $c\text{-}CH_2OCH_2$ is measured with very low mole fractions (<8 ppm). Excluding unimolecular reactions has in this case only a minor effect on the species profiles and the THP conversion. This is in line with the work performed on THF [11].

The dominant hydrogen abstraction from THP gives the THP-YL-2 radical. The latter reacts by the C–O bond β -scission leading to the formation of the $R52C_5H_9OA$ radical. Less important consumption channels of the THP-YL-2 radical are C–C bond β -scission yielding the $R53C_5H_9OEZ$ radical and C–H bond β -scission producing 3,4-DHP. $R52C_5H_9OA$ and $R53C_5H_9OEZ$ radicals, acyclic C_5H_9O isomers, decompose by β -scission of the C–C bonds to produce C_2H_4 and C_3H_5O radicals. The $R52C_5H_9OA$ radical can also isomerize to $R58C_5H_9OA$ and $R59C_5H_9OK$, which decompose to produce C_2H_5 plus C_2H_3CHO and CO plus nC_4H_9 , respectively. Decomposition of THP-YL-2 is a major source for the formation of C_2H_4 , CO , $1\text{-}C_4H_8$, and C_2H_3CHO . The present model reproduces the mole fraction profiles of these abundant species quite well, as shown in Figs. 7, 9, 12 (for the burner-stabilized flame data) and Figs. 2, 3, 5 (the pyrolysis data), demonstrating that their main formation channels are well taken into account.

The THP-YL-3 radical reacts by β -scission of C–O, C–C, or C–H bonds to produce $R51C_5H_9OZ$, $R50C_5H_9OEZ$ radicals, or dihydropyrans (3,4-DHP and 3,6-DHP), respectively. The $R51C_5H_9OZ$ radical decomposes into formaldehyde and the $C_4H_7\text{-}1$ radical ($CH_2=CH\text{-}CH_2\text{-}CH_2$). This pathway is important for HCHO formation. As shown in Figs. 11 and 5, the HCHO profiles are quite well reproduced by the model. The $R50C_5H_9OEZ$ radical reacts via C–O bond

β -scission or via isomerization to produce C_2H_4 plus $R56C_3H_5OZ$ or $R55C_5H_9OEX$ radical, respectively. These last two radicals form C_2H_3CHO ; however, this is only a minor pathway compared to reaction channels starting from THP-YL-2.

The THP-YL-4 radical is mainly consumed by C–C bond β -scissions to form $R54C_5H_9OEZ$ subsequently decomposing to HCHO and the $C_4H_7\text{-}1$ radical, similar to $R51C_5H_9OZ$. The $C_4H_7\text{-}1$ radical decomposes into C_2H_4 plus C_2H_3 or H plus $1,3\text{-}C_4H_6$. This pathway contributes significantly to the formation of $1,3\text{-}C_4H_6$, the most abundant C_4 intermediate.

Rate of production analysis further shows that significant amounts of 3,4-DHP and 3,6-DHP are formed from three fuel radicals. These molecules are mainly consumed by H-abstractions and partially by retro-Diels–Alder reactions. The contribution of the retro-Diels–Alder reaction to the 3,4-DHP and 3,6-DHP consumptions increases with temperature. The good prediction of this species with the current mechanism presents a clear improvement over the model of Labbe et al. [10] (Fig. 12). A large part of the C_2H_2 production results from the reaction $C_2H_3(+M)=C_2H_2+H(+M)$. In the early stages of reaction, C_2H_3 is mainly produced by the THP reactions via the formation of the $C_4H_7\text{-}1$, $R62C_3H_5OEZ$, and $R56C_3H_5OZ$ radicals (Fig. 16). The C_2H_3 radical can recombine with the CH_3 radical contributing to C_3H_6 formation. C_3H_8 and $1\text{-}C_5H_{10}$ are mainly produced by the reactions $C_2H_5+CH_3=C_3H_8$ and $1\text{-}C_5H_{10}=C_3H_5Y+C_2H_5$, respectively. As shown in Fig. 16, C_2H_5 can be formed directly from THP via the nC_4H_9 radical. Hence, the formation of C_3H_6 , C_3H_8 , and $1\text{-}C_5H_{10}$ is affected by radicals resulting from the fuel consumption. They were detected as abundant C_3 (Fig. 8) and C_5 (Fig. 10) intermediates, and their mole fraction profiles are quite well predicted by the present model. CH_3CHO is mainly formed by the reactions: $1,3\text{-}C_4H_6+OH=CH_3CHO+C_2H_3$, $CH_2CHO+H=CH_3CHO$,



In general, in the fuel-rich THP flame the same reactions are obviously important as those in the combustion of THP with some differences in their respective importance. For instance, when the

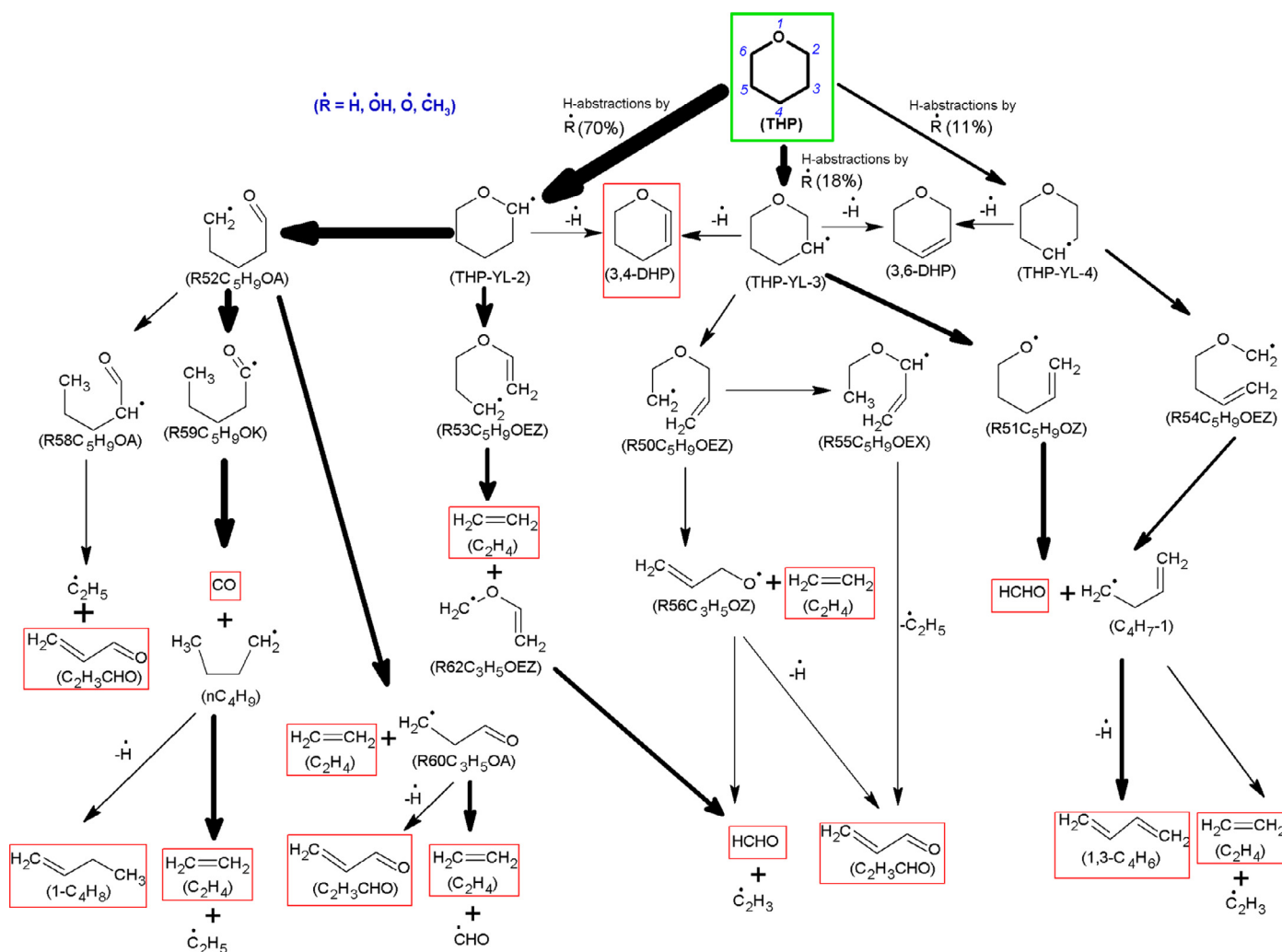


Fig. 16. Rate-of-production analysis for the consumption of THP in the stoichiometric flame for a distance of 1.35 mm from the burner corresponding to a temperature of 1053 K and 70% conversion of THP. The size of the arrows is proportional to the relative rates of consumption of a given species. Species in the squares have been experimentally detected.

equivalence ratio decreases, the importance of reactions involving oxygenated radicals such as OH radicals or O-atoms is enhanced, while that of reactions involving non-oxygenated radicals such as H-atoms or CH₃ radicals is reduced.

6.3. Sensitivity analyses

To identify the reactions which are sensitive for the fuel consumption at these conditions, a sensitivity analysis for THP was performed at two different positions in the reactor (at 260 and 500 mm from the inlet of reactor) corresponding to 13% (Fig. 17a) and 50% (Fig. 17b) fuel conversion. Figure 17 shows that the reactions yielding small radicals enhance the fuel consumption, such as $C_2H_5(+M) = C_2H_4 + H(+M)$, while an opposite trend is noted for the combinations of small radicals, such as $2C_2H_5 \Rightarrow nC_4H_{10}$. At the conditions of this sensitivity analysis, the most sensitive reaction of Table 3 is the C₂–C₃ bond fission, which produces C[•]CCCOC (Fig. 17) via the formation of a diradical (B16, see Fig. 15). The model predicts that C[•]CCCOC is converted to C₃H₆ and c-CH₂OCH₂, which can decompose into two reactive radicals CH₃ and CHO, increasing the global reactivity. C[•]CCCOC can also decompose into C₃H₅Y + HCHO + CH₃ (where C₃H₅Y is the allyl radical), again supplying radicals to the system. Early in the reactor (Fig. 17a), H-abstractions by H-atom involving THP-YL-2, THP-YL-3, and THP-YL-4 are not sensitive, but at later stages (Fig. 17b), they become more sensitive as the H atom concentration increases.

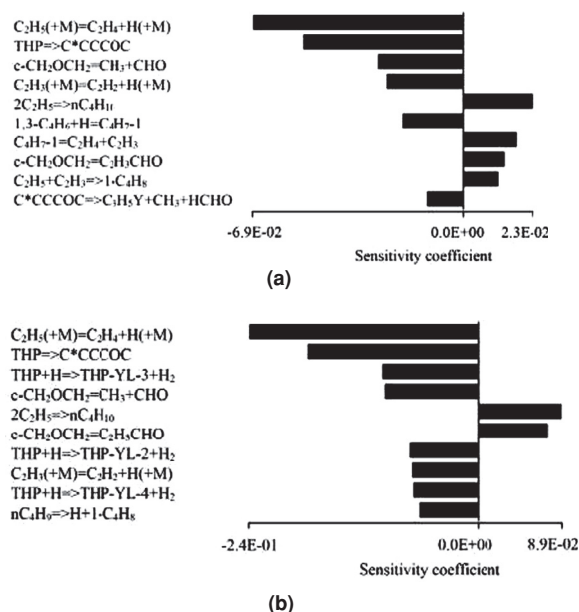


Fig. 17. Sensitivity analysis on THP for pyrolysis conditions 1053 K, 170 kPa at around 260 mm (a) and 500 mm (b) from the inlet of reactor, corresponding to ~13% and ~50% fuel conversion, respectively. A negative coefficient corresponds to a reaction which enhances the consumption of THP and vice versa.

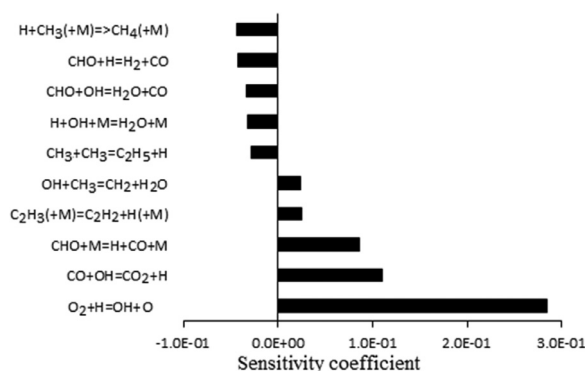


Fig. 18. Sensitivity analysis of computed laminar burning velocities for a stoichiometric THP-air mixture at 1.0 atm and 298 K of initial conditions. A negative coefficient corresponds to a reaction which reduces the computed laminar burning velocity and vice versa.

To identify the reactions which control the predictions of adiabatic laminar burning velocities of THP in air, a sensitivity analysis is carried out for an initial temperature of 298 K at $\varphi = 1.0$ and presented in Fig. 18 for the ten most sensitive reactions. None of the reactions in Table 3 are listed among these ten reactions. The kinetics of small species control the adiabatic laminar burning velocities. Especially the chain branching reaction of O₂ with H, producing OH and O (O₂+H=OH+O), enhances burning velocities. In contrast, the recombination of H and OH with other radicals, producing stable products, inhibits flame propagation. This trend has been observed in most studies previously published in the literature [11,38,75]. As a test, additional simulations were performed using the rate constant of the reaction O₂+H=OH+O proposed recently by Hong et al. [76] ($k = 10.40 \times 10^{13} [\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}] \exp(-15,286 [\text{cal mol}^{-1}]/RT)$ instead of that ($k = 9.75 \times 10^{13} [\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}] \exp(-14,844 [\text{cal mol}^{-1}]/RT)$ proposed by Baulch et al. [77] (used in the present model). As shown in Fig. 13 (thin lines), the modified model predicts the experimental data slightly better but it still overpredicts the flame velocities in lean and stoichiometric mixtures.

The effect of the unimolecular initiations involving C–C and C–O scission of THP and the decomposition of the resulting products on the simulated adiabatic laminar burning velocities is very limited, as shown in Fig. 13 (dotted line at 298 K). These reactions do not appear in the sensitivity analysis (Fig. 18). A similar trend was also pointed out in a previous study on THF [11].

For ignition delay times, a comparison between simulations for selected mixtures with and without the unimolecular initiations involving C–C and C–O scissions (reactions 1–5 of Table 3) is presented in Fig. 14c. The ignition delay times change slightly without these scission channels. To better understand which reactions influence the predictions of auto-ignition delay times, a sensitivity analysis with respect to the OH* radical concentration, i.e. auto-ignition delay time, is performed and presented in Fig. 19. The sensitivity analysis corresponds to $\varphi = 1.0$ (1.0% THP/7.0% O₂/92.0% Ar), a temperature of 1452 K, and a pressure of 8.98 atm. As shown in Fig. 19 the reactions of small species are very sensitive for the OH* formation, which implies that, similarly to burning velocity analysis, the modeling auto-ignition delay time is controlled significantly by the kinetics of small species. Some of the primary reactions of Table 3 appear in this analysis, i.e. the reaction pathway THP→CCCCCHO→CH₂CHO+nC₃H₇ and H-abstraction from C₂ of THP by a H-atom. The first reaction sequence enhances the formation of OH* because it yields two radicals (CH₂CHO and nC₃H₇), whereas the last one has an inhibiting effect because of H-atom consumption, lowering the amount of H-atoms available to perform chain branching reactions with O₂. Since unimolecular initiations, with considerable uncertainty in their rate constants as mentioned earlier, appear among the ten most sensitive reactions of the sensitivity analysis, an improvement in their kinetic

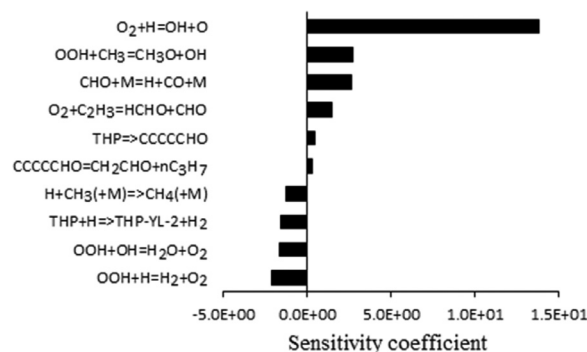


Fig. 19. Sensitivity analysis on the OH* radical for ignition delay time at $\varphi = 1.0$ (1.0% THP/7.0% O₂/92.0% Ar), 1452 K, and 8.98 atm.

data using further theoretical studies could improve the agreement between model and experiment.

7. Summary and conclusion

The pyrolysis and combustion of tetrahydropyran (THP) has been investigated experimentally and numerically in a tubular flow reactor, in flat premixed flames, and in a shock tube. Over 20 species have been identified and quantified in these different experiments. Ethylene was found to be the most abundant intermediate during the pyrolysis/combustion of THP, analogous to other saturated fuels such as ethanol and THF. 1,3-Butadiene, formaldehyde, and acrolein are other important intermediates. For the yields of aromatic species strong differences were observed between combustion and pyrolysis conditions. Under pyrolysis conditions they were measured in non-negligible amounts, while very low concentrations under flame conditions were observed, sometimes lower than the detection limit of the gas chromatograph. The yields of CO by elimination reactions and the formation of C₁–C₃ intermediates in the THP flames are quite similar to those obtained for THF flames.

To gain further insight in the pyrolysis and combustion chemistry of THP a new kinetic model was constructed using EXGAS. This model is able to reproduce the experimental data of this study and those of Labbe et al. [10]. For comparison simulations with the models of the literature were also presented and the main differences were identified. The present model predicts quite well a large part of pyrolysis and combustion data obtained in this study. Nevertheless further work would be useful for improving experimental results by using more accurate calibration factors and refining the kinetic mechanism including rate coefficients obtained from higher level calculations.

Reaction path analysis shows that THP is largely consumed by H-abstractions in both pyrolysis and combustion conditions. The unimolecular initiations involving C–C and C–O bond scission play a very minor role under the studied combustion conditions, while their kinetics strongly affect pyrolysis modeling results. This suggests that theoretical studies regarding unimolecular fuel decomposition would be extremely useful to improve understanding of the different initiation pathways and their kinetics.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.combustflame.2015.07.030](https://doi.org/10.1016/j.combustflame.2015.07.030).

References

- [1] A. Osment, L. Catoire, P. Escot Bocanegra, I. Gökalp, B. Thollas, J.A. Kozinski, *Combust. Flame* 157 (2010) 1230–1234.
- [2] R.J. van Putten, J.C. van der Waal, E. de Jong, C.B. Rasrendra, H.J. Heeres, J.G. de Vries, *Chem. Rev.* 113 (2013) 1499–1597.
- [3] G.W. Huber, S. Iborra, A. Corma, *Chem. Rev.* 106 (2006) 4044–4098.
- [4] W.R. Leppard, *SAE Trans.* 100 (1991) 589–604.
- [5] J. Bünger, J. Kralh, K. Baum, O. Schröder, M. Müller, G. Westphal, P. Ruhnuau, T.G. Schulz, *E. Arch. Toxicology* 74 (2000) 490–498.
- [6] K. Kohse-Höinghaus, P. Oßwald, T.A. Cool, T. Kasper, N. Hansen, F. Qi, C.K. Westbrook, P.R. Westmoreland, *Angew. Chem. Int. Ed.* 49 (2010) 3572–3597.
- [7] P. Dagaut, M. Reuillon, M. Cathonnet, *Combust. Sci. Technol.* 103 (1994) 315–336.
- [8] O. Herbinet, S. Bax, P.-A. Glaude, V. Carre, F. Battin-Leclerc, *Fuel* 90 (2011) 528–535.
- [9] P. Dagaut, M. McGuinness, J.M. Simmie, M. Cathonnet, *Combust. Sci. Technol.* 129 (1997) 1–16.
- [10] N.J. Labbe, V. Seshadri, T. Kasper, N. Hansen, P. Oßwald, P.R. Westmoreland, *Proc. Combust. Inst.* 34 (2013) 259–267.
- [11] L.-S. Tran, M. Verdichio, F. Monge, R.C. Martin, R. Bounaceur, B. Sirjean, P.-A. Glaude, M.U. Alzueta, F. Battin-Leclerc, *Combust. Flame* 162 (2015) 1899–1918.
- [12] Z. Serinyel, O. Herbinet, O. Frotier, P. Dirrenberger, V. Warth, P.-A. Glaude, F. Battin-Leclerc, *Combust. Flame* 160 (2013) 2319–2332.
- [13] M.R. Harper, K.M. Van Geem, S.P. Pyl, G.B. Marin, W.H. Green, *Combust. Flame* 158 (2011) 16–41.
- [14] M. Djokic, H.-H. Carstensen, K.M. Van Geem, G.B. Marin, *Proc. Combust. Inst.* 34 (2013) 251–258.
- [15] M.R. Djokic, K.M. Van Geem, C. Cavallotti, A. Frassoldati, E. Ranzi, G.B. Marin, *Combust. Flame* 161 (2014) 2739–2751.
- [16] S.P. Pyl, K.M. Van Geem, P. Puimege, M.K. Sabbe, M.-F. Reyniers, G.B. Marin, *Energy* 43 (2012) 146–160.
- [17] S.P. Pyl, C.M. Schietekat, K.M. Van Geem, M.-F. Reyniers, J. Vercammen, J. Beens, G.B. Marin, *J. Chromatogr. A* 1218 (2011) 3217–3223.
- [18] K.M. Van Geem, S.P. Pyl, M.-F. Reyniers, J. Vercammen, J. Beens, G.B. Marin, *J. Chromatogr. A* 1217 (2010) 6623–6633.
- [19] J. Beens, H. Boelens, R. Tijssen, J. Blomberg, *J. High Resolut. Chromatogr.* 21 (1998) 47–54.
- [20] E. Pousse, Z.Y. Tian, P.A. Glaude, R. Fournet, F. Battin-Leclerc, *Combust. Flame* 157 (2010) 1236–1260.
- [21] L.-S. Tran, P.-A. Glaude, R. Fournet, F. Battin-Leclerc, *Energy Fuels* 27 (2013) 2226–2245.
- [22] D. Liu, C. Togbé, L.-S. Tran, D. Felsmann, P. Oßwald, P. Nau, J. Koppmann, A. Lackner, P.-A. Glaude, B. Sirjean, R. Fournet, F. Battin-Leclerc, K. Kohse-Höinghaus, *Combust. Flame* 161 (2014) 748–765.
- [23] J.H. Kent, *Combust. Flame* 14 (1970) 279–282.
- [24] U. Bonne, T. Grewer, H.G. Wagner, *Z. Phys. Chem* 26 (1960) 93–110.
- [25] L.H.P. De Goey, A. Van Maaren, R.M. Quax, *Combust. Sci. Technol.* 92 (1993) 201–207.
- [26] P. Dirrenberger, H. Le Gall, R. Bounaceur, O. Herbinet, P.-A. Glaude, A. Konnov, F. Battin-Leclerc, *Energy Fuels* 25 (2011) 3875–3884.
- [27] P. Dirrenberger, P.-A. Glaude, R. Bounaceur, H. Le Gall, A. Pires da Cruz, A. Konnov, F. Battin-Leclerc, *Fuel* 115 (2014) 162–169.
- [28] B. Sirjean, F. Buda, H. Hakka, P.A. Glaude, R. Fournet, V. Warth, F. Battin-Leclerc, M. Ruiz-Lopez, *Proc. Combust. Inst.* 31 (2007) 277–284.
- [29] H. Bennadji, L. Coniglio, F. Billaud, R. Bounaceur, V. Warth, P.-A. Glaude, F. Battin-Leclerc, *Int. J. Chem. Kinet.* 43 (2011) 204.
- [30] J.A. Montgomery Jr., M.J. Frisch, J.W. Ochterski, G.A. Petersson, *J. Chem. Phys.* 110 (1999) 2822–2827.
- [31] Gaussian 09, Revision B.01, M.J. Frisch, G.W. Trucks, H.B. Schlegel, G.E. Scuseria, M.A. Robb, J.R. Cheeseman, G. Scalmani, V. Barone, B. Mennucci, G.A. Petersson, H. Nakatsuji, M. Caricato, X. Li, H.P. Hratchian, A.F. Izmaylov, J. Bloino, G. Zheng, J.L. Sonnenberg, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, J.A. Montgomery, Jr., J.E. Peralta, F. Ogliaro, M. Bearpark, J.J. Heyd, E. Brothers, K.N. Kudin, V.N. Staroverov, R. Kobayashi, J. Normand, K. Raghavachari, A. Rendell, J.C. Burant, S.S. Iyengar, J. Tomasi, M. Cossi, N. Rega, J.M. Millam, M. Klene, J.E. Knox, J.B. Cross, V. Bakken, C. Adamo, J. Jaramillo, R. Comperts, R.E. Stratmann, O. Yazyev, A.J. Austin, R. Cammi, C. Pomelli, J.W. Ochterski, R.L. Martin, K. Morokuma, V.G. Zakrzewski, G.A. Voth, P. Salvador, J.J. Dannenberg, S. Dapprich, A.D. Daniels, Ö. Farkas, J.B. Foresman, J.V. Ortiz, J. Cioslowski, D.J. Fox, Gaussian, Inc., Wallingford CT, 2009.
- [32] A.L.L. East, L. Radom, *J. Chem. Phys.* 106 (1997) 6655–6674.
- [33] CHEMKIN-PRO, Reaction Design Inc., San Diego, CA, 2010.
- [34] R.J. Kee, F.M. Rupley, J.A. Miller, Sandia Laboratories Report no. SAND-89-8009B, 1993.
- [35] A.T. Hartlieb, B. Atakan, K. Kohse-Höinghaus, *Combust. Flame* 121 (2000) 610–624.
- [36] U. Struckmeier, P. Obwald, T. Kasper, L. Böling, M. Heusing, M. Köhler, A. Brockhinke, K. Kohse-Höinghaus, *Z. Phys. Chem.* 223 (2009) 503–537.
- [37] A. Bhargava, P.R. Westmoreland, *Combust. Flame* 113 (1998) 333–347.
- [38] K.P. Somers, J.M. Simmie, F. Gillespie, C. Conroy, G. Black, W.K. Metcalfe, F. Battin-Leclerc, P. Dirrenberger, O. Herbinet, P.-A. Glaude, P. Dagaut, C. Togbé, K. Yasunaga, R.X. Fernandes, C. Lee, R. Tripathi, H.J. Curran, *Combust. Flame* 160 (2013) 2291–2318.
- [39] B. Sirjean, R. Fournet, P.-A. Glaude, F. Battin-Leclerc, W. Wang, M.A. Oehlschlaeger, *J. Phys. Chem. A* 117 (2013) 1371–1392.
- [40] P. Barbé, F. Battin-Leclerc, G.M. Côme, *J. Chim. Phys.* 92 (1995) 1666–1692.
- [41] V. Warth, N. Stef, P.A. Glaude, F. Battin-Leclerc, G. Scacchi, G.M. Côme, *Combust. Flame* 114 (1998) 81–102.
- [42] F. Buda, R. Bounaceur, V. Warth, P.A. Glaude, R. Fournet, F. Battin-Leclerc, *Combust. Flame* 142 (2005) 170–186.
- [43] B. Heyberger, N. Belmekki, V. Conraud, P.A. Glaude, R. Fournet, F. Battin-Leclerc, *Int. J. Chem. Kin.* 36 (2002) 666–677.
- [44] B. Husson, M. Ferrari, O. Herbinet, S.S. Ahmed, P.A. Glaude, F. Battin-Leclerc, *Proc. Combust. Inst.* 34 (2013) 325–333.
- [45] W. Tsang, R.F. Hampson, *J. Phys. Chem. Ref. Data* 15 (1986) 1087–1279.
- [46] D.L. Baulch, C.T. Bowman, C.J. Cobos, R.A. Cox, T. Just, J.A. Kerr, M.J. Pilling, D. Stocker, J. Troe, W. Tsang, R.W. Walker, J. Warnatz, *J. Phys. Chem. Ref. Data* 34 (2005) 757–1397.
- [47] R. Fournet, J.C. Bauge, F. Battin-Leclerc, *Int. J. Chem. Kinet.* 31 (1999) 361–379.
- [48] N. Belmekki, P.A. Glaude, I. Da Costa, R. Fournet, F. Battin-Leclerc, *Int. J. Chem. Kinet.* 34 (2002) 172–183.
- [49] H.A. Gueniche, J. Biet, P.A. Glaude, R. Fournet, F. Battin-Leclerc, *Fuel* 88 (2009) 1388–1393.
- [50] W. Li, M.E. Law, P.R. Westmoreland, T. Kasper, N. Hansen, K. Kohse-Höinghaus, *Combust. Flame* 158 (2011) 2077–2089.
- [51] M. Verdichio, B. Sirjean, L.-S. Tran, P.-A. Glaude, F. Battin-Leclerc, *Proc. Combust. Inst.* 35 (2015) 533–541.
- [52] B.E. Poling, J.M. Prausnitz, O’Connell, *The Properties of Gases and Liquids*, fifth ed., The McGraw-Hill Companies, 2001 ISBN:0070116822.
- [53] B. Sirjean, P.A. Glaude, M.F. Ruiz-Lopez, R. Fournet, *J. Phys. Chem. A* 110 (2006) 12693–12704.
- [54] A.M. Dean, J.W. Bozzelli, *Combustion Chemistry of Nitrogen*, in: Gas-Phase Combustion Chemistry, in: W.C. Gardiner (Ed.), Springer, New York, 2000.
- [55] J. Zhang, L. Pan, J. Mo, J. Gong, Z. Huang, C.K. Law, *Combust. Flame* 160 (2013) 1541–1549.
- [56] P.S. Veloo, P. Dagaut, C. Togbé, G. Dayma, S.M. Sarathy, C.K. Westbrook, F.N. Egol-fopoulos, *Combust. Flame* 160 (2013) 1609–1626.
- [57] C. Muller, V. Michel, G. Scacchi, G.M. Come, *J. Chem. Phys.* 92 (1995) 1154.
- [58] S.W. Benson, *Thermochemical Kinetics*, 2nd ed., Wiley, New York, 1976.
- [59] Burcat database <http://burcat.technion.ac.il/dir/>.
- [60] A.S. Pell, G. Pilcher, *Trans. Faraday Soc* 61 (1965) 71–77.
- [61] O.V. Dorofeeva, *Thermochim. Acta* 200 (1992) 121–150.
- [62] H. Wang, M. Frenklach, *Combust. Flame* 96 (1994) 163–170.
- [63] N.A. Slavinskaya, P. Frank, *Combust. Flame* 156 (2009) 1705–1722.
- [64] V. Chernov, M.J. Thomson, S.B. Dworkin, N.A. Slavinskaya, U. Riedel, *Combust. Flame* 161 (2014) 592–601.
- [65] A. Lifshitz, M. Bidani, S. Bidani, *J. Phys. Chem.* 90 (1986) 3422–3429.
- [66] J.H. Kiefer, K.S. Gupte, L.B. Harding, S.J. Klippenstein, *J. Phys. Chem. A* 113 (2009) 13570–13583.
- [67] B. Sirjean, P.-A. Glaude, M.F. Ruiz-Lopez, R. Fournet, *J. Phys. Chem. A* 110 (2006) 12693–12704.
- [68] L.-S. Tran, C. Togbé, D. Liu, D. Felsmann, P. Oßwald, P.-A. Glaude, R. Fournet, B. Sirjean, F. Battin-Leclerc, K. Kohse-Höinghaus, *Combust. Flame* 161 (2013) 766–779.
- [69] Z. Tian, Y. Yuan, R. Fournet, P.-A. Glaude, B. Sirjean, F. Battin-Leclerc, K. Zhang, F. Qi, *Combust. Flame* 158 (2011) 756–773.
- [70] L.-S. Tran, B. Sirjean, P.-A. Glaude, K. Kohse-Höinghaus, F. Battin-Leclerc, *Proc. Combust. Inst.* 35 (2015) 1735–1743.
- [71] C. Togbé, L.-S. Tran, D. Liu, D. Felsmann, P. Oßwald, P.-A. Glaude, B. Sirjean, R. Fournet, F. Battin-Leclerc, K. Kohse-Höinghaus, *Combust. Flame* 161 (2013) 780–797.
- [72] S.G. Davis, C.K. Law, *Combust. Sci. Technol.* 140 (1998) 427–449.
- [73] S.S. Merchant, E.F. Zanoelo, R.L. Speth, M.R. Harper, K.M. Van Geem, W.H. Green, *Combust. Flame* 160 (2013) 1907–1929.
- [74] M.K. Sabbe, K.M. Van Geem, M.-F. Reyniers, G.B. Marin, *AIChE J.* 57 (2011) 482–496.
- [75] E. Ranzi, A. Frassoldati, R. Grana, A. Cuoci, T. Faravelli, A.P. Kelley, C.K. Law, *Prog. Energy Combust. Sci.* 38 (2012) 468–501.
- [76] Z. Hong, D.F. Davidson, E.A. Barbour, R.K. Hanson, *Proc. Combust. Inst.* 33 (2011) 309–316.
- [77] D.L. Baulch, C.J. Cobos, R.A. Cox, P. Frank, G. Hayman, Th. Just, J.A. Kerr, T. Murrells, M.J. Pilling, J. Troe, R.W. Walker, J. Warnatz, *J. Phys. Chem. Ref. Data* 23 (1994) 847–1033.